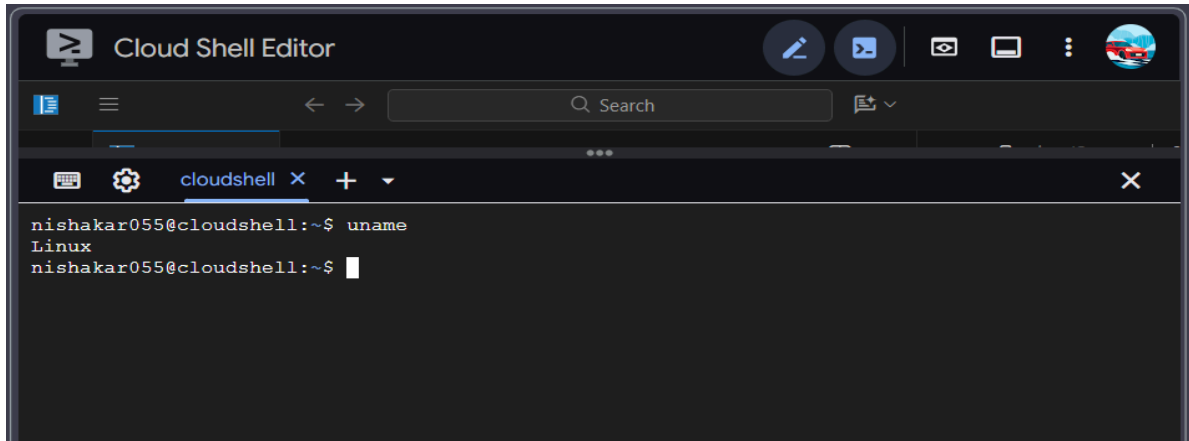


Linux commands

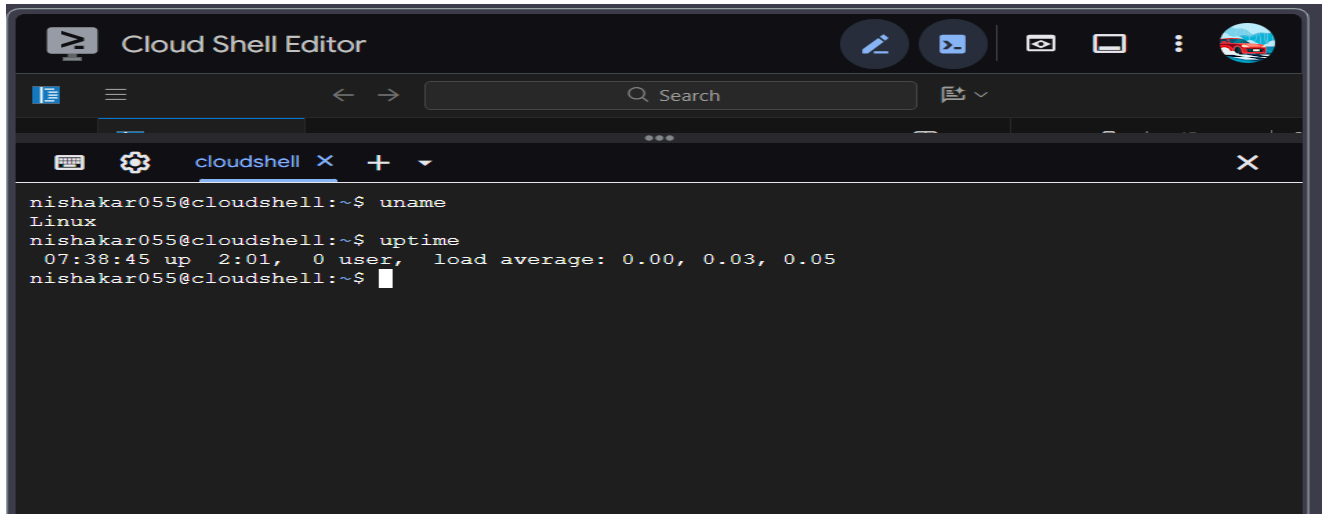
1.uname



The screenshot shows the Cloud Shell Editor interface. The terminal window displays the command 'uname' and its output 'Linux'. The prompt is 'nishakar055@cloudshell:~\$'.

```
nishakar055@cloudshell:~$ uname
Linux
nishakar055@cloudshell:~$
```

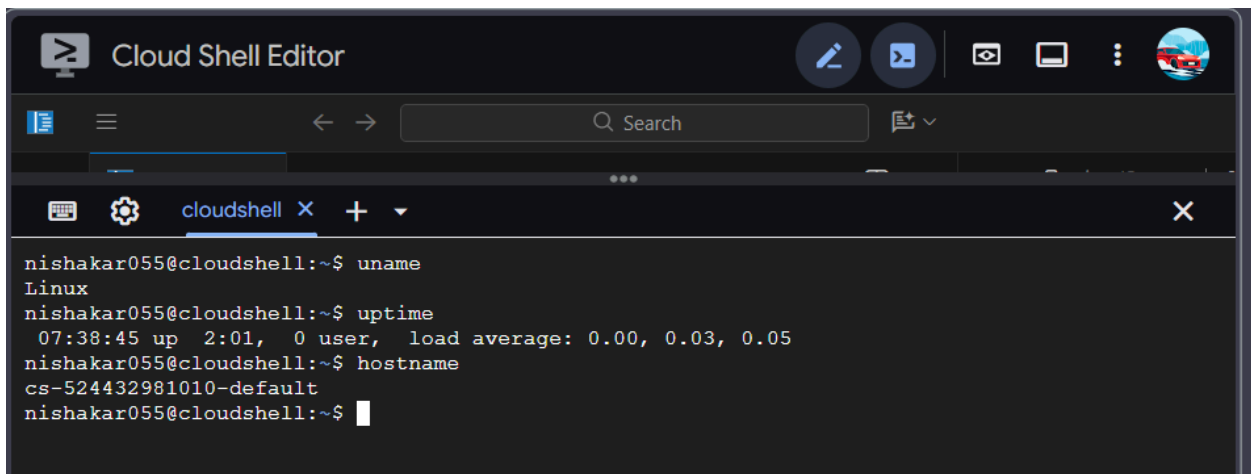
2.uptime



The screenshot shows the Cloud Shell Editor interface. The terminal window displays the command 'uptime' and its output. The prompt is 'nishakar055@cloudshell:~\$'.

```
nishakar055@cloudshell:~$ uname
Linux
nishakar055@cloudshell:~$ uptime
 07:38:45 up  2:01,  0 user,  load average: 0.00, 0.03, 0.05
nishakar055@cloudshell:~$
```

3.hostname



The screenshot shows the Cloud Shell Editor interface. The terminal window displays the command 'hostname' and its output 'cs-524432981010-default'. The prompt is 'nishakar055@cloudshell:~\$'.

```
nishakar055@cloudshell:~$ uname
Linux
nishakar055@cloudshell:~$ uptime
 07:38:45 up  2:01,  0 user,  load average: 0.00, 0.03, 0.05
nishakar055@cloudshell:~$ hostname
cs-524432981010-default
nishakar055@cloudshell:~$
```

4.free

```
Cloud Shell Editor

nishakar055@cloudshell:~$ uname
Linux
nishakar055@cloudshell:~$ uptime
07:38:45 up 2:01, 0 user, load average: 0.00, 0.03, 0.05
nishakar055@cloudshell:~$ hostname
cs-524432981010-default
nishakar055@cloudshell:~$ free
              total        used        free      shared  buff/cache   available
Mem:      16382844      3898300     10881856        1136       1884080       12484544
Swap:            0              0              0
```

5.top

```
Cloud Shell Editor

%Cpu(s):  0.4 us,  0.4 sy,  0.0 ni, 99.1 id,  0.0 wa,  0.0 hi,  0.1 si,  0.0 st
MiB Mem : 15998.9 total, 10622.9 free,  3810.8 used,  1839.9 buff/cache
MiB Swap:  0.0 total,  0.0 free,  0.0 used. 12188.0 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
  486 root        20   0 1867468   46520  32616 S   0.3   0.3   0:06.02 containerd
  535 nishaka+    20   0  11.3g 109072  51228 S   0.3   0.7   0:10.83 node
 1944 nishaka+    20   0  42.4g 762156  62276 S   0.3   4.7   0:36.02 node
    1 root        20   0   4324    3316  3032 S   0.0   0.0   0:00.08 bash
    9 syslog    20   0  222140    3808  2452 S   0.0   0.0   0:00.05 rsyslogd
   25 root        20   0  38768   27988 10880 S   0.0   0.2   0:01.19 python
   26 root        20   0   6196    2620  2380 S   0.0   0.0   0:00.00 logger
   67 root       10 -10  12024    4300  3184 S   0.0   0.0   0:00.05 sshd
  363 root       20   0 2096096  85764  62100 S   0.0   0.5   0:00.97 dockerd
  437 root       20   0 1234524  12612  8208 S   0.0   0.1   0:01.83 editor-proxy
  438 root       20   0   16328    6420  5496 S   0.0   0.0   0:00.01 sudo
  452 root       20   0 1226356   2832  1976 S   0.0   0.0   0:00.00 tmux-agent
  510 root       20   0    9196    4068  3692 S   0.0   0.0   0:00.00 runuser
  511 nishaka+    20   0   4324    3376  3112 S   0.0   0.0   0:00.00 sh
  632 nishaka+   10 -10   7568    3136  2100 S   0.0   0.0   0:00.58 tmux
  633 nishaka+   10 -10   8800    5960  4084 S   0.0   0.0   0:00.30 bash
  771 root       20   0   2696    1476  1380 S   0.0   0.0   0:00.00 sleep
 1549 nishaka+    20   0  31.4g 217032  60412 S   0.0   1.3   0:08.25 node
 1897 nishaka+   10 -10   6076    2728  2384 T   0.0   0.0   0:00.01 less
 1915 nishaka+   10 -10   6076    2664  2324 T   0.0   0.0   0:00.00 less
 1940 nishaka+   10 -10  24704   13140  7864 S   0.0   0.1   0:00.23 vi
 1955 nishaka+   20   0 1048516  62360  47840 S   0.0   0.4   0:00.52 node
 1975 nishaka+   20   0 1016536  60496  46280 S   0.0   0.4   0:00.42 node
 1983 nishaka+   20   0  31.4g 216736  60696 S   0.0   1.3   0:08.28 node
 1997 nishaka+   20   0 1853348  616244  49100 S   0.0   3.8   0:09.23 node
 2004 nishaka+   20   0 1951792  72780  52204 S   0.0   0.4   0:00.69 cloudcode_cli
 2042 nishaka+   10 -10   8800    5808  3932 S   0.0   0.0   0:00.14 bash
 2444 root       10 -10  13996    9784  8404 S   0.0   0.1   0:00.02 sshd
 2446 root       10 -10  13996    9856  8480 S   0.0   0.1   0:00.02 sshd
 2448 nishaka+   10 -10  14256    6328  4672 S   0.0   0.0   0:00.00 sshd
 2449 nishaka+   10 -10  14256    6488  4832 S   0.0   0.0   0:00.12 sshd
 2450 nishaka+   10 -10   7340    3664  3416 S   0.0   0.0   0:00.00 bash
 2453 nishaka+   10 -10   7604    4460  3852 S   0.0   0.0   0:00.00 bash

[4]+  Stopped                  top
nishakar055@cloudshell:~$
```

6.htop

```
0[|||] 1.3% Tasks: 47, 164 thr, 0 kthr; 1 running
1[|||] 1.3% Load averages: 0.00 0.02 0.05
2[|||] 0.0% Uptime: 02:02:40
3[|||] 1.3%
Mem[|||||] 3.45G/15.6G
Swp[|] 0K/0K

Main I/O
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
448 root 20 0 1205M 12612 8208 S 0.7 0.1 0:00.43 /google/devshell/editor/
492 root 20 0 1823M 46520 32616 S 0.7 0.3 0:01.04 containerd --config /var
2645 nishakar05 10 -10 8448 4764 3680 R 0.7 0.0 0:00.05 htop/sbin/rsyslogd
10 syslog 20 0 216M 3808 2452 S 0.0 0.0 0:00.02 /usr/sbin/rsyslogd
11 syslog 20 0 216M 3808 2452 S 0.0 0.0 0:00.00 /usr/sbin/rsyslogd
12 syslog 20 0 216M 3808 2452 S 0.0 0.0 0:00.02 /usr/sbin/rsyslogd
25 root 20 0 38768 27988 10880 S 0.0 0.2 0:01.19 /usr/bin/python /usr/bin
26 root 20 0 6196 2620 2380 S 0.0 0.0 0:00.00 logger -t supervisord
67 root 10 -10 12024 4300 3184 S 0.0 0.0 0:00.05 sshd: /usr/sbin/sshd -p
363 root 20 0 2046M 85764 62100 S 0.0 0.5 0:00.21 /usr/bin/dockerd -p /var
411 root 20 0 2046M 85764 62100 S 0.0 0.5 0:00.13 /usr/bin/dockerd -p /var
412 root 20 0 2046M 85764 62100 S 0.0 0.5 0:00.09 /usr/bin/dockerd -p /var
413 root 20 0 2046M 85764 62100 S 0.0 0.5 0:00.00 /usr/bin/dockerd -p /var
414 root 20 0 2046M 85764 62100 S 0.0 0.5 0:00.14 /usr/bin/dockerd -p /var
415 root 20 0 2046M 85764 62100 S 0.0 0.5 0:00.00 /usr/bin/dockerd -p /var
435 root 20 0 2046M 85764 62100 S 0.0 0.5 0:00.00 /usr/bin/dockerd -p /var
437 root 20 0 1205M 12612 8208 S 0.0 0.1 0:00.31 /google/devshell/editor/
438 root 20 0 16328 6420 5496 S 0.0 0.0 0:00.01 sudo /google/devshell/tm
448 root 20 0 1205M 12612 8208 S 0.0 0.1 0:00.42 /google/devshell/editor/

449 root 20 0 1205M 12612 8208 S 0.0 0.1 0:00.34 /google/devshell/editor/
450 root 20 0 1205M 12612 8208 S 0.0 0.1 0:00.20 /google/devshell/editor/
451 root 20 0 1205M 12612 8208 S 0.0 0.1 0:00.20 /google/devshell/editor/
452 root 20 0 1197M 2832 1976 S 0.0 0.0 0:00.00 /google/devshell/tmux-ag
461 root 20 0 1197M 2832 1976 S 0.0 0.0 0:00.00 /google/devshell/tmux-ag
463 root 20 0 1197M 2832 1976 S 0.0 0.0 0:00.00 /google/devshell/tmux-ag
464 root 20 0 1197M 2832 1976 S 0.0 0.0 0:00.00 /google/devshell/tmux-ag
465 root 20 0 1197M 2832 1976 S 0.0 0.0 0:00.00 /google/devshell/tmux-ag
482 root 20 0 1205M 12612 8208 S 0.0 0.1 0:00.28 /google/devshell/editor/

F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice F8Nice F9Kill F10Quit
```

7.lsb_release

```
nishakar055@cloudshell:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.4 LTS
Release:        24.04
Codename:       noble
nishakar055@cloudshell:~$
```

8.arch

```
Cloud Shell Editor

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't show again Dismiss

nishakar055@cloudshell:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.4 LTS
Release:       24.04
Codename:      noble
nishakar055@cloudshell:~$ arch
x86_64
nishakar055@cloudshell:~$
```

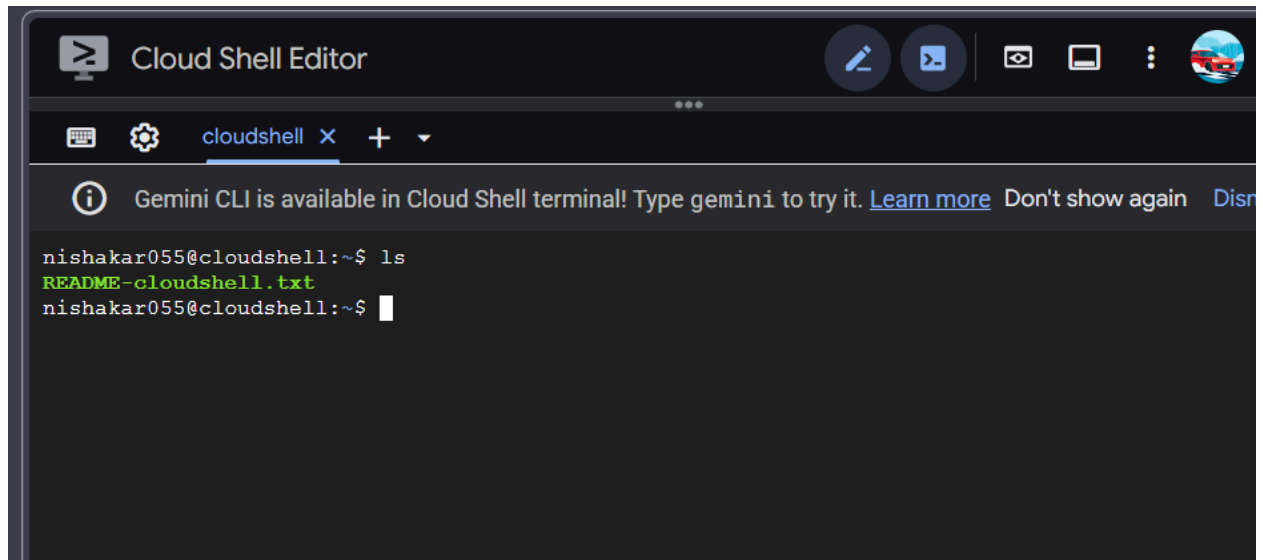
9.dmesg

```
Cloud Shell Editor

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't show again Dismiss

filter/nft_limit.ko" pid=1535 cmdline="/sbin/modprobe -q -- nft-expr-limit"
[12322.086875] cni0: port 2(veth1102c950) entered blocking state
[12322.092918] cni0: port 2(veth1102c950) entered disabled state
[12322.092960] veth1102c950: entered allmulticast mode
[12322.104019] veth1102c950: entered promiscuous mode
[12322.119560] cni0: port 2(veth1102c950) entered blocking state
[12322.125642] cni0: port 2(veth1102c950) entered forwarding state
[12324.472727] scsi 0:0:2:0: Direct-Access Google PersistentDisk 1 PQ: 0 ANSI: 6
[12324.482900] sd 0:0:2:0: [sdb] 10485760 512-byte logical blocks: (5.37 GB/5.00 GiB)
[12324.490661] sd 0:0:2:0: [sdb] 4096-byte physical blocks
[12324.496361] sd 0:0:2:0: [sdb] Write Protect is off
[12324.501442] sd 0:0:2:0: [sdb] Mode Sense: 1f 00 00 08
[12324.501649] sd 0:0:2:0: [sdb] Write cache: enabled, read cache: enabled, doesn't support DPC
r FUA
[12324.584256] sdb: sdb1
[12324.587073] sd 0:0:2:0: [sdb] Attached SCSI disk
[12324.984081] cni0: port 2(veth1102c950) entered disabled state
[12324.991515] veth1102c950 (unregistering): left allmulticast mode
[12324.998155] veth1102c950 (unregistering): left promiscuous mode
[12325.005842] cni0: port 2(veth1102c950) entered disabled state
[12325.172806] cni0: port 2(veth5e5bbb0e) entered blocking state
[12325.178855] cni0: port 2(veth5e5bbb0e) entered disabled state
[12325.178887] veth5e5bbb0e: entered allmulticast mode
[12325.190029] veth5e5bbb0e: entered promiscuous mode
[12325.196348] cni0: port 2(veth5e5bbb0e) entered blocking state
[12325.203518] cni0: port 2(veth5e5bbb0e) entered forwarding state
[12325.439650] EXT4-fs (sdb1): mounted filesystem b9f0e455-8c03-483b-a0d6-629a47561fb9 r/w with
rdered data mode. Quota mode: none.
[12326.363417] LoadPin: kernel-module pinning-excluded obj="/lib/modules/6.6.137+/kernel/net/ig
/netfilter/iptable_nat.ko" pid=3978 cmdline="/sbin/modprobe -q -- iptable_nat"
[12326.395643] LoadPin: kernel-module pinning-excluded obj="/lib/modules/6.6.137+/kernel/net/ne
ink/netlink_diag.ko" pid=3985 cmdline="/sbin/modprobe -q -- net-pf-16-proto-4-type-16"
[12329.095400] LoadPin: kernel-module pinning-excluded obj="/lib/modules/6.6.137+/kernel/net/ne
ilter/ipset/ip_set.ko" pid=4528 cmdline="/sbin/modprobe -q -- ipt_set"
[12329.118186] LoadPin: kernel-module pinning-excluded obj="/lib/modules/6.6.137+/kernel/net/ne
ilter/xt_set.ko" pid=4528 cmdline="/sbin/modprobe -q -- ipt_set"
[12329.149380] LoadPin: kernel-module pinning-excluded obj="/lib/modules/6.6.137+/kernel/net/ip
/netfilter/ip6table_nat.ko" pid=4534 cmdline="/sbin/modprobe -q -- ip6table_nat"
[12379.966047] icmp: detected local route for 10.0.1.108 during ICMP sending, src 216.239.36.17
nishakar055@cloudshell:~$
```

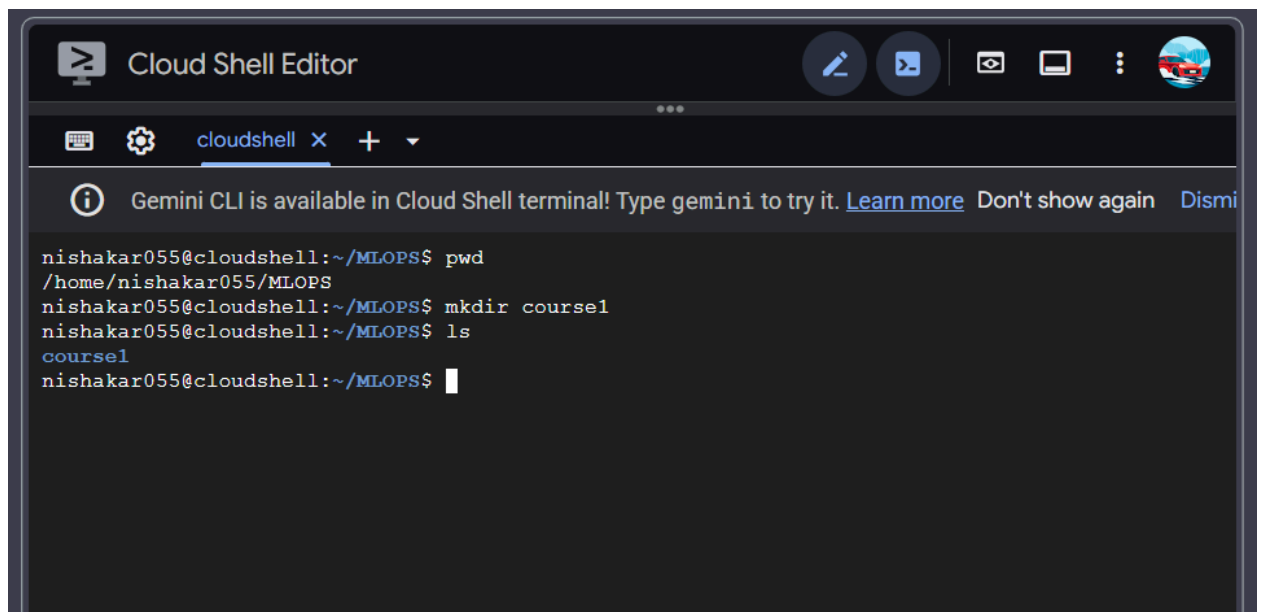
10.ls



The screenshot shows the Cloud Shell Editor interface. At the top, there's a title bar with a terminal icon and the text "Cloud Shell Editor". Below it, a tab labeled "cloudshell" is active. A notification banner at the top states: "Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#) Don't show again Dismiss". The terminal window shows the user "nishakar055@cloudshell" at the prompt "nishakar055@cloudshell:~\$". They have entered the command "ls", and the output is "README-cloudshell.txt".

```
nishakar055@cloudshell:~$ ls
README-cloudshell.txt
nishakar055@cloudshell:~$
```

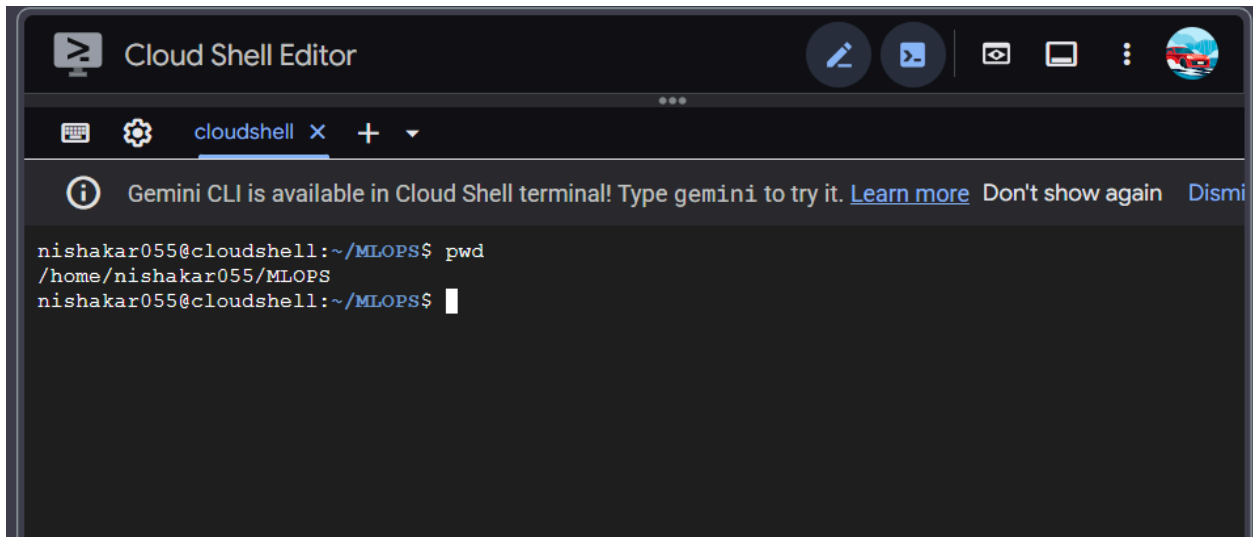
11.mkdir



The screenshot shows the Cloud Shell Editor interface. At the top, there's a title bar with a terminal icon and the text "Cloud Shell Editor". Below it, a tab labeled "cloudshell" is active. A notification banner at the top states: "Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#) Don't show again Dismiss". The terminal window shows the user "nishakar055@cloudshell" at the prompt "nishakar055@cloudshell:~/MLOPS\$". They have entered the command "pwd", and the output is "/home/nishakar055/MLOPS". They have then entered the command "mkdir course1", and the output is "course1". Finally, they have entered the command "ls", and the output is "course1".

```
nishakar055@cloudshell:~/MLOPS$ pwd
/home/nishakar055/MLOPS
nishakar055@cloudshell:~/MLOPS$ mkdir course1
nishakar055@cloudshell:~/MLOPS$ ls
course1
nishakar055@cloudshell:~/MLOPS$
```

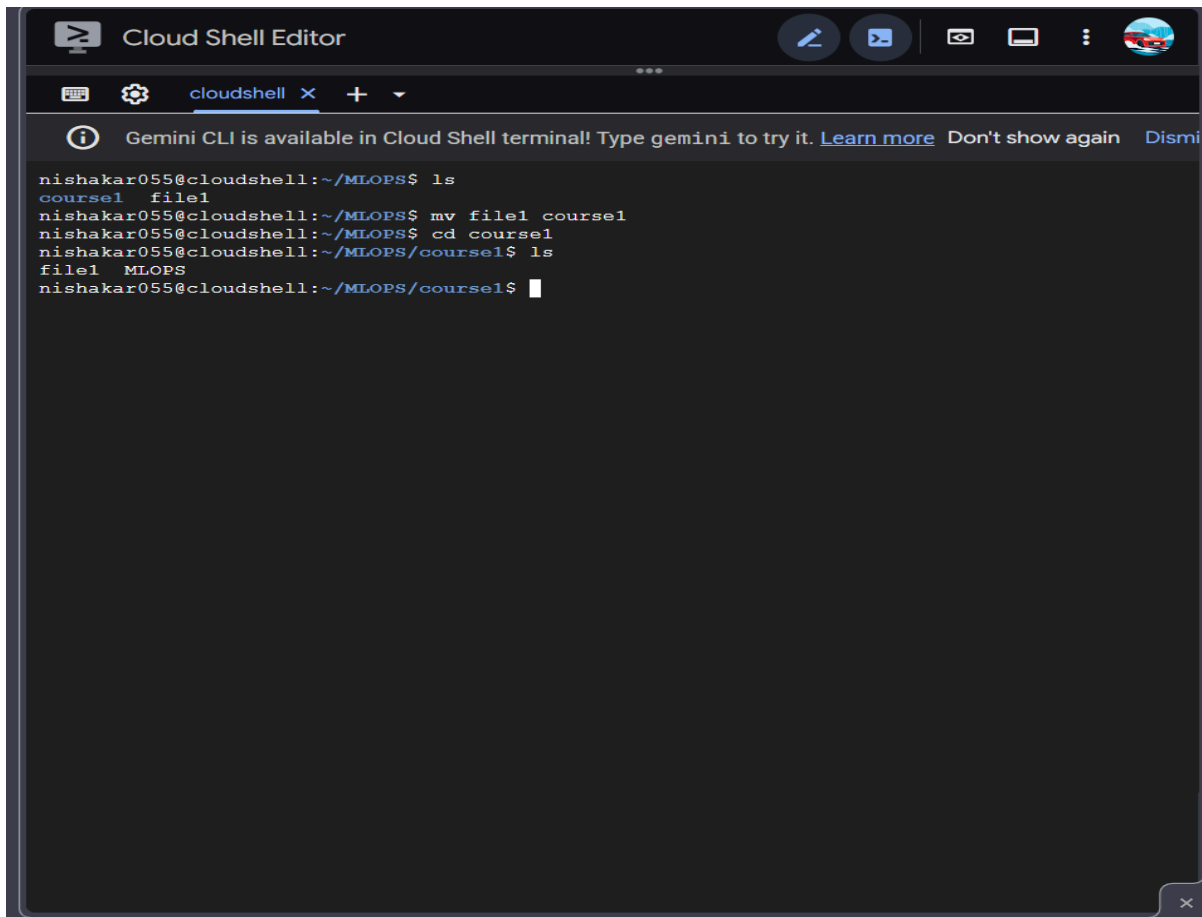
12.pwd



The screenshot shows the Cloud Shell Editor interface. At the top, there's a title bar with a terminal icon and the text "Cloud Shell Editor". Below it, a tab labeled "cloudshell" is active. A notification banner at the top states: "Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#) Don't show again Dismiss". The terminal window shows the following commands and output:

```
nishakar055@cloudshell:~/MLOPS$ pwd
/home/nishakar055/MLOPS
nishakar055@cloudshell:~/MLOPS$
```

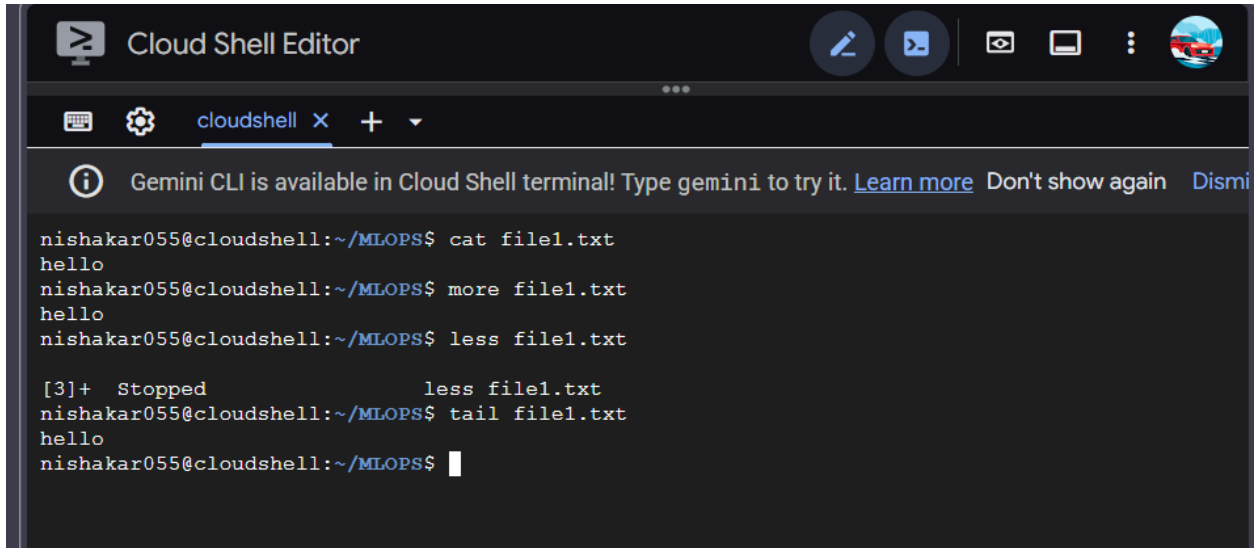
13.cd



The screenshot shows the Cloud Shell Editor interface. At the top, there's a title bar with a terminal icon and the text "Cloud Shell Editor". Below it, a tab labeled "cloudshell" is active. A notification banner at the top states: "Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#) Don't show again Dismiss". The terminal window shows the following commands and output:

```
nishakar055@cloudshell:~/MLOPS$ ls
course1  file1
nishakar055@cloudshell:~/MLOPS$ mv file1 course1
nishakar055@cloudshell:~/MLOPS$ cd course1
nishakar055@cloudshell:~/MLOPS/course1$ ls
file1  MLOPS
nishakar055@cloudshell:~/MLOPS/course1$
```

14.tail

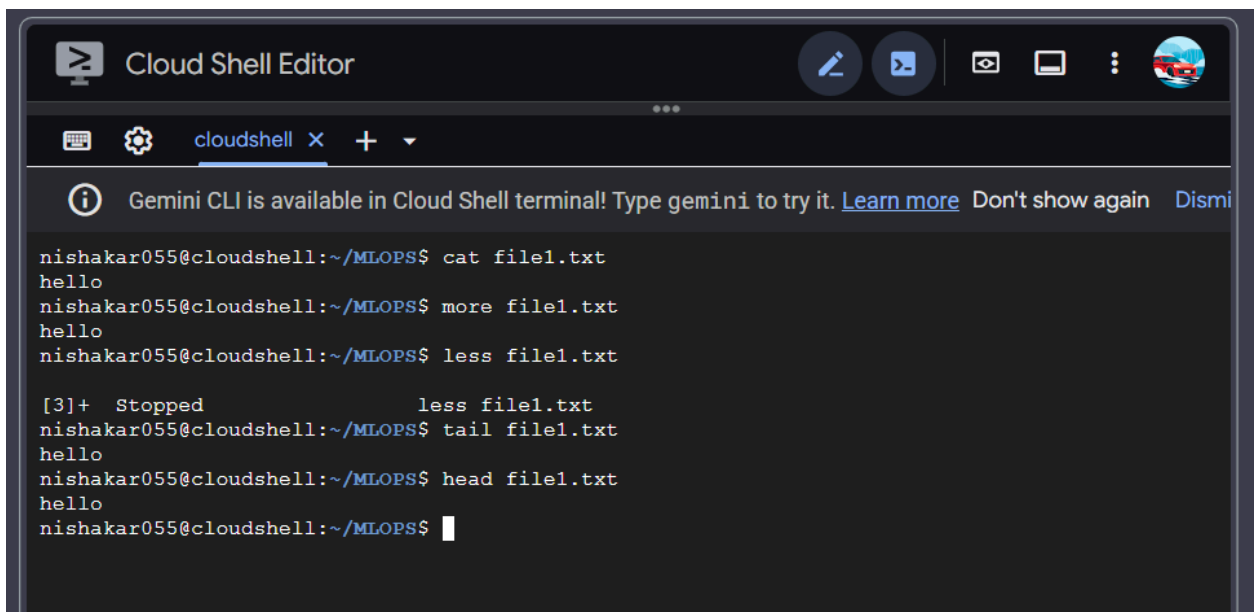


The screenshot shows the Cloud Shell Editor interface. At the top, there's a title bar with a terminal icon and the text "Cloud Shell Editor". Below it, a toolbar contains icons for editing, saving, and other functions. A tab labeled "cloudshell" is active. A notification banner at the top states: "Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#) Don't show again Dismiss". The terminal window shows the following commands and output:

```
nishakar055@cloudshell:~/MLOPS$ cat file1.txt
hello
nishakar055@cloudshell:~/MLOPS$ more file1.txt
hello
nishakar055@cloudshell:~/MLOPS$ less file1.txt

[3]+  Stopped                  less file1.txt
nishakar055@cloudshell:~/MLOPS$ tail file1.txt
hello
nishakar055@cloudshell:~/MLOPS$
```

15.head

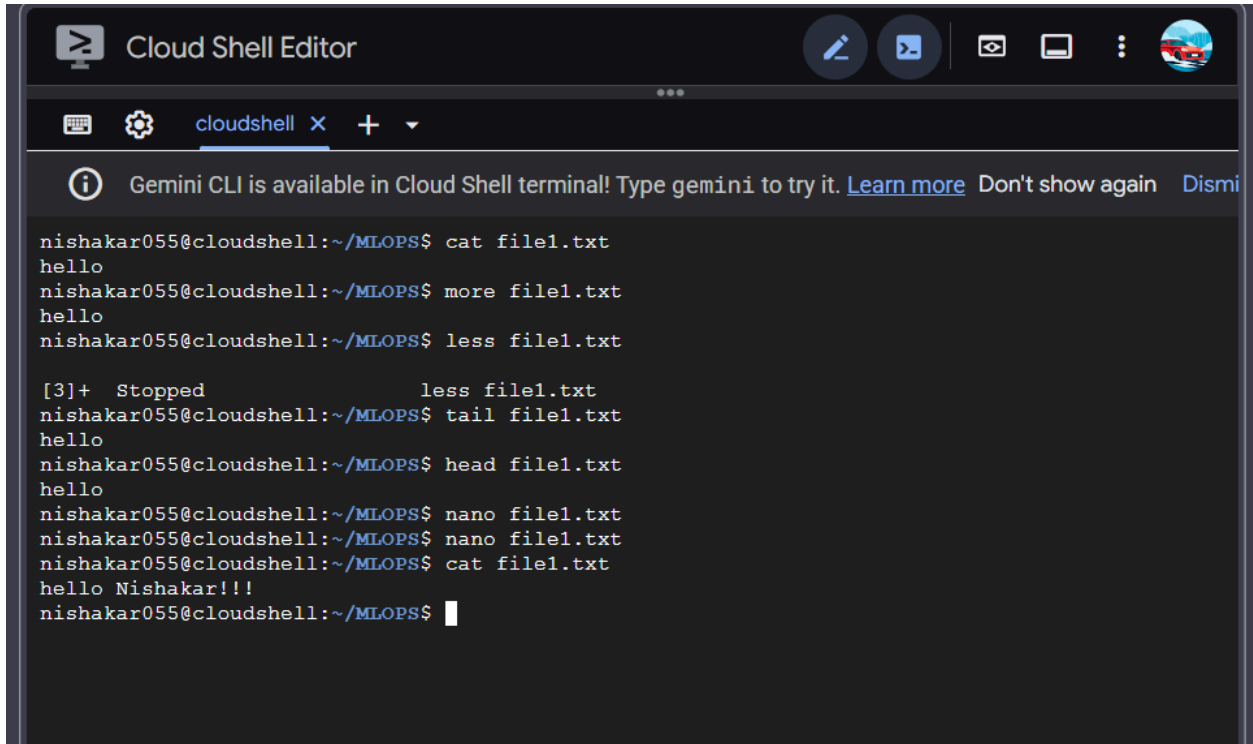


The screenshot shows the Cloud Shell Editor interface, similar to the previous one. The terminal window shows the following commands and output:

```
nishakar055@cloudshell:~/MLOPS$ cat file1.txt
hello
nishakar055@cloudshell:~/MLOPS$ more file1.txt
hello
nishakar055@cloudshell:~/MLOPS$ less file1.txt

[3]+  Stopped                  less file1.txt
nishakar055@cloudshell:~/MLOPS$ tail file1.txt
hello
nishakar055@cloudshell:~/MLOPS$ head file1.txt
hello
nishakar055@cloudshell:~/MLOPS$
```

16.nano

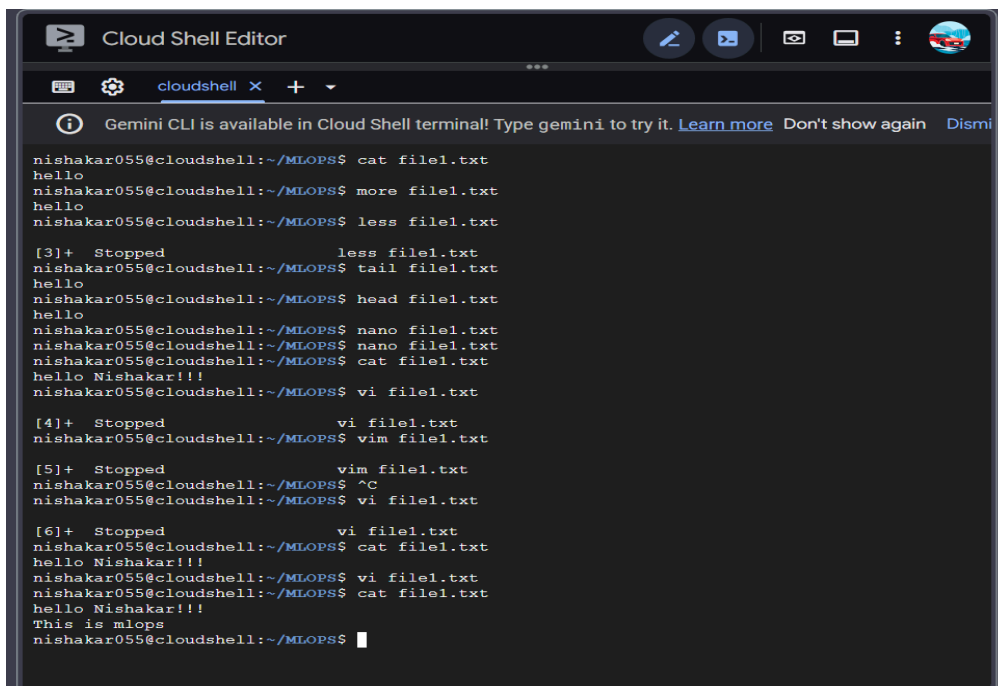


The screenshot shows a Cloud Shell Editor window with a terminal. The terminal output is as follows:

```
nishakar055@cloudshell:~/MLOPS$ cat file1.txt
hello
nishakar055@cloudshell:~/MLOPS$ more file1.txt
hello
nishakar055@cloudshell:~/MLOPS$ less file1.txt

[3]+  Stopped                  less file1.txt
nishakar055@cloudshell:~/MLOPS$ tail file1.txt
hello
nishakar055@cloudshell:~/MLOPS$ head file1.txt
hello
nishakar055@cloudshell:~/MLOPS$ nano file1.txt
nishakar055@cloudshell:~/MLOPS$ nano file1.txt
nishakar055@cloudshell:~/MLOPS$ cat file1.txt
hello Nishakar!!!
nishakar055@cloudshell:~/MLOPS$
```

17.vi



The screenshot shows a Cloud Shell Editor window with a terminal. The terminal output is as follows:

```
nishakar055@cloudshell:~/MLOPS$ cat file1.txt
hello
nishakar055@cloudshell:~/MLOPS$ more file1.txt
hello
nishakar055@cloudshell:~/MLOPS$ less file1.txt

[3]+  Stopped                  less file1.txt
nishakar055@cloudshell:~/MLOPS$ tail file1.txt
hello
nishakar055@cloudshell:~/MLOPS$ head file1.txt
hello
nishakar055@cloudshell:~/MLOPS$ nano file1.txt
nishakar055@cloudshell:~/MLOPS$ nano file1.txt
nishakar055@cloudshell:~/MLOPS$ cat file1.txt
hello Nishakar!!!
nishakar055@cloudshell:~/MLOPS$ vi file1.txt

[4]+  Stopped                  vi file1.txt
nishakar055@cloudshell:~/MLOPS$ vim file1.txt

[5]+  Stopped                  vim file1.txt
nishakar055@cloudshell:~/MLOPS$ ^C
nishakar055@cloudshell:~/MLOPS$ vi file1.txt

[6]+  Stopped                  vi file1.txt
nishakar055@cloudshell:~/MLOPS$ cat file1.txt
hello Nishakar!!!
nishakar055@cloudshell:~/MLOPS$ vi file1.txt
nishakar055@cloudshell:~/MLOPS$ cat file1.txt
hello Nishakar!!!
This is mlops
nishakar055@cloudshell:~/MLOPS$
```


18.awk

```
Cloud Shell Editor

nishakar055@cloudshell:~$ ls
MLOPS  README-cloudshell.txt
nishakar055@cloudshell:~$ cd MLOPS
nishakar055@cloudshell:~/MLOPS$ ls
course1 file1.txt file1.txtxo txt
nishakar055@cloudshell:~/MLOPS$ cd txt
-bash: cd: txt: Not a directory
nishakar055@cloudshell:~/MLOPS$ cat file1.txtxo
hello Nishakar!!!

nishakar055@cloudshell:~/MLOPS$ cat file1.txt
hello Nishakar!!!
This is mlops
nishakar055@cloudshell:~/MLOPS$ nano file1.txt
nishakar055@cloudshell:~/MLOPS$ cat file1.txt
hello Nishakar!!!
This is mlops
nishakar 100
nish 200
nish2 300

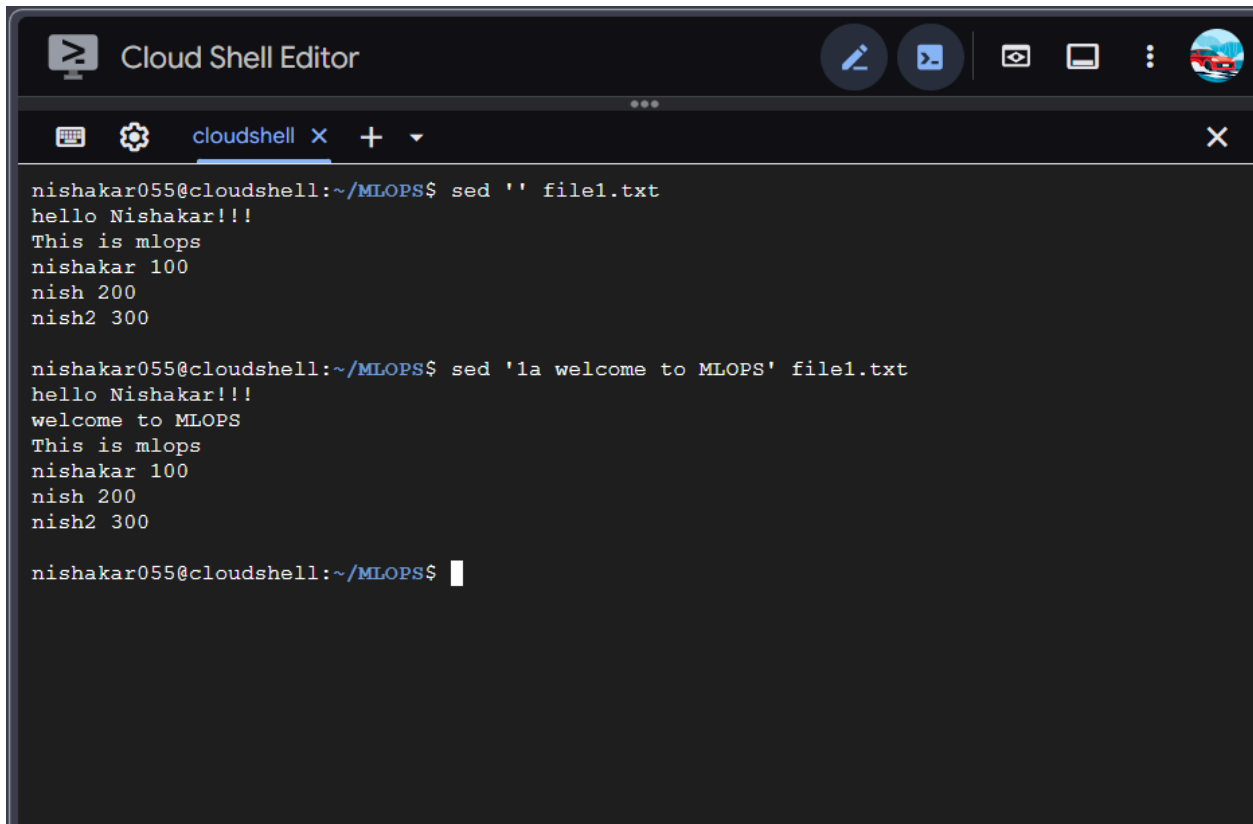
nishakar055@cloudshell:~/MLOPS$ awk {'print $1'} file1.txt
hello
This
nishakar
nish
nish2

nishakar055@cloudshell:~/MLOPS$ awk {'print $2'} file1.txt
Nishakar!!!
is
100
200
300

nishakar055@cloudshell:~/MLOPS$ awk {'print $3'} file1.txt
mlops

nishakar055@cloudshell:~/MLOPS$
```

19.sed



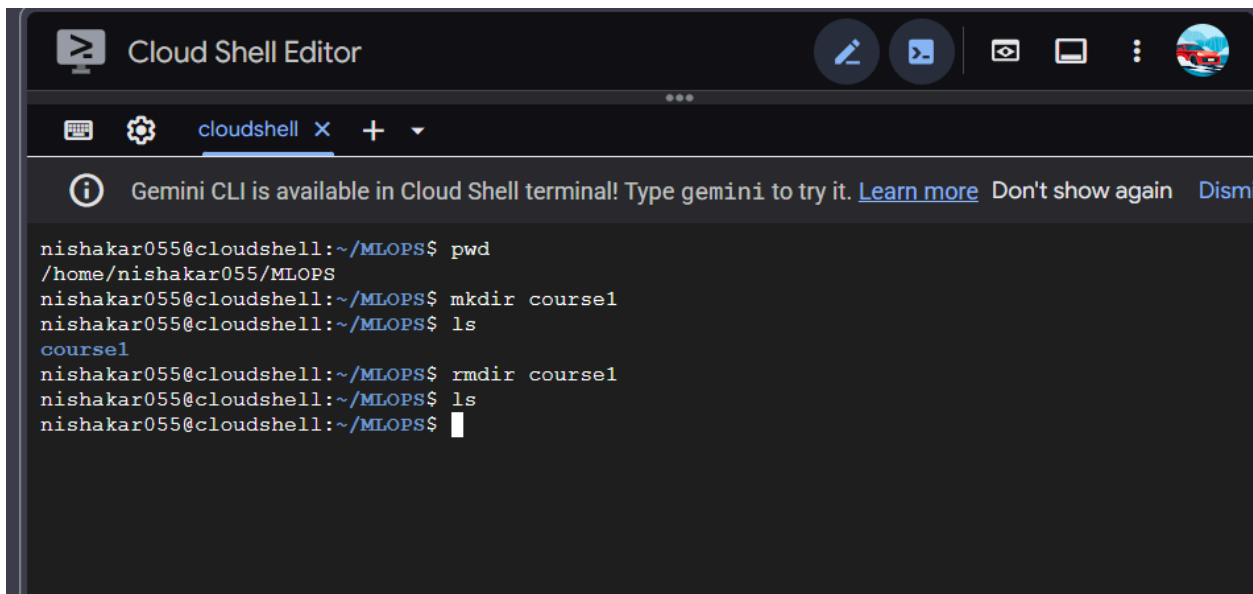
The screenshot shows a Cloud Shell Editor window with a terminal. The user is in the directory ~/MLOPS. They run the command `sed '' file1.txt`, which prints the contents of file1.txt: `hello Nishakar!!!`, `This is mlops`, `nishakar 100`, `nish 200`, and `nish2 300`. Then, they run `sed '1a welcome to MLOPS' file1.txt`, which prints the same content but with `welcome to MLOPS` added as the first line. The terminal prompt is `nishakar055@cloudshell:~/MLOPS$`.

```
nishakar055@cloudshell:~/MLOPS$ sed '' file1.txt
hello Nishakar!!!
This is mlops
nishakar 100
nish 200
nish2 300

nishakar055@cloudshell:~/MLOPS$ sed '1a welcome to MLOPS' file1.txt
hello Nishakar!!!
welcome to MLOPS
This is mlops
nishakar 100
nish 200
nish2 300

nishakar055@cloudshell:~/MLOPS$
```

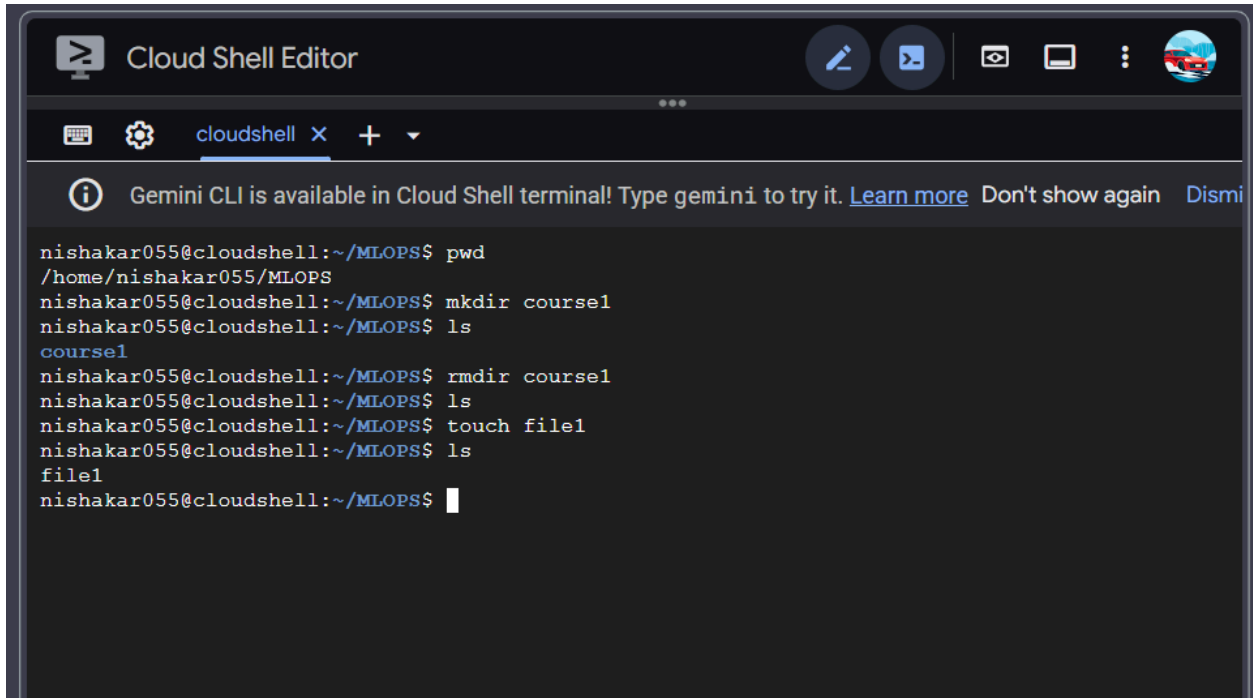
20.rmdir



The screenshot shows a Cloud Shell Editor window with a terminal. A notification banner at the top says "Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#) Don't show again Dismiss". The user is in the directory ~/MLOPS. They run `pwd`, which outputs `/home/nishakar055/MLOPS`. Then, they run `mkdir course1`. Next, they run `ls`, which outputs `course1`. Then, they run `rmdir course1`. Finally, they run `ls` again, which outputs nothing, indicating the directory has been removed. The terminal prompt is `nishakar055@cloudshell:~/MLOPS$`.

```
nishakar055@cloudshell:~/MLOPS$ pwd
/home/nishakar055/MLOPS
nishakar055@cloudshell:~/MLOPS$ mkdir course1
nishakar055@cloudshell:~/MLOPS$ ls
course1
nishakar055@cloudshell:~/MLOPS$ rmdir course1
nishakar055@cloudshell:~/MLOPS$ ls
nishakar055@cloudshell:~/MLOPS$
```

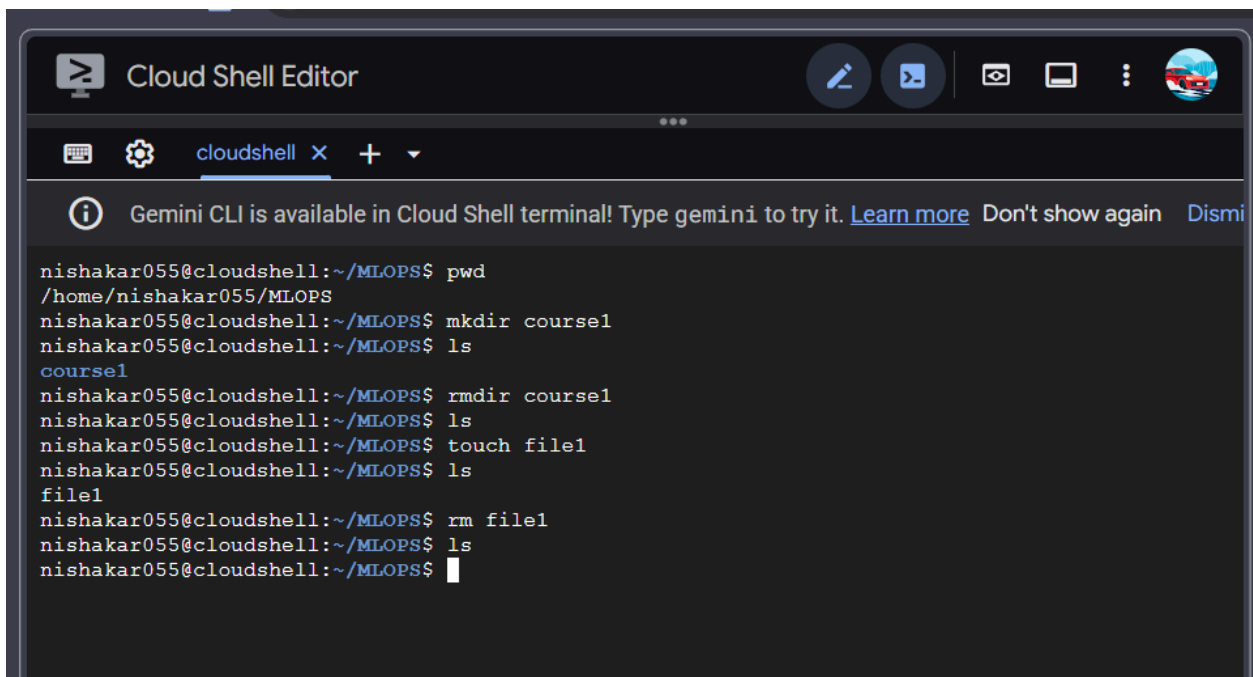
21.touch



The image shows a Cloud Shell Editor terminal window. The title bar reads "Cloud Shell Editor". Below the title bar, there is a tab labeled "cloudshell" with a close button and a plus sign. A notification banner at the top states: "Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#) Don't show again Dismiss". The terminal content shows the following commands and output:

```
nishakar055@cloudshell:~/MLOPS$ pwd
/home/nishakar055/MLOPS
nishakar055@cloudshell:~/MLOPS$ mkdir course1
nishakar055@cloudshell:~/MLOPS$ ls
course1
nishakar055@cloudshell:~/MLOPS$ rmdir course1
nishakar055@cloudshell:~/MLOPS$ ls
nishakar055@cloudshell:~/MLOPS$ touch file1
nishakar055@cloudshell:~/MLOPS$ ls
file1
nishakar055@cloudshell:~/MLOPS$
```

22.rm



The image shows a Cloud Shell Editor terminal window, similar to the one above. The title bar reads "Cloud Shell Editor". Below the title bar, there is a tab labeled "cloudshell" with a close button and a plus sign. A notification banner at the top states: "Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#) Don't show again Dismiss". The terminal content shows the following commands and output:

```
nishakar055@cloudshell:~/MLOPS$ pwd
/home/nishakar055/MLOPS
nishakar055@cloudshell:~/MLOPS$ mkdir course1
nishakar055@cloudshell:~/MLOPS$ ls
course1
nishakar055@cloudshell:~/MLOPS$ rmdir course1
nishakar055@cloudshell:~/MLOPS$ ls
nishakar055@cloudshell:~/MLOPS$ touch file1
nishakar055@cloudshell:~/MLOPS$ ls
file1
nishakar055@cloudshell:~/MLOPS$ rm file1
nishakar055@cloudshell:~/MLOPS$ ls
nishakar055@cloudshell:~/MLOPS$
```

23.mv

```
Cloud Shell Editor

cloudshell x + v

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't show again Dismiss

nishakar055@cloudshell:~/MLOPS$ ls
course1  file1
nishakar055@cloudshell:~/MLOPS$ mv file1 course1
nishakar055@cloudshell:~/MLOPS$ cd course1
nishakar055@cloudshell:~/MLOPS/course1$ ls
file1  MLOPS
nishakar055@cloudshell:~/MLOPS/course1$
```

24.In

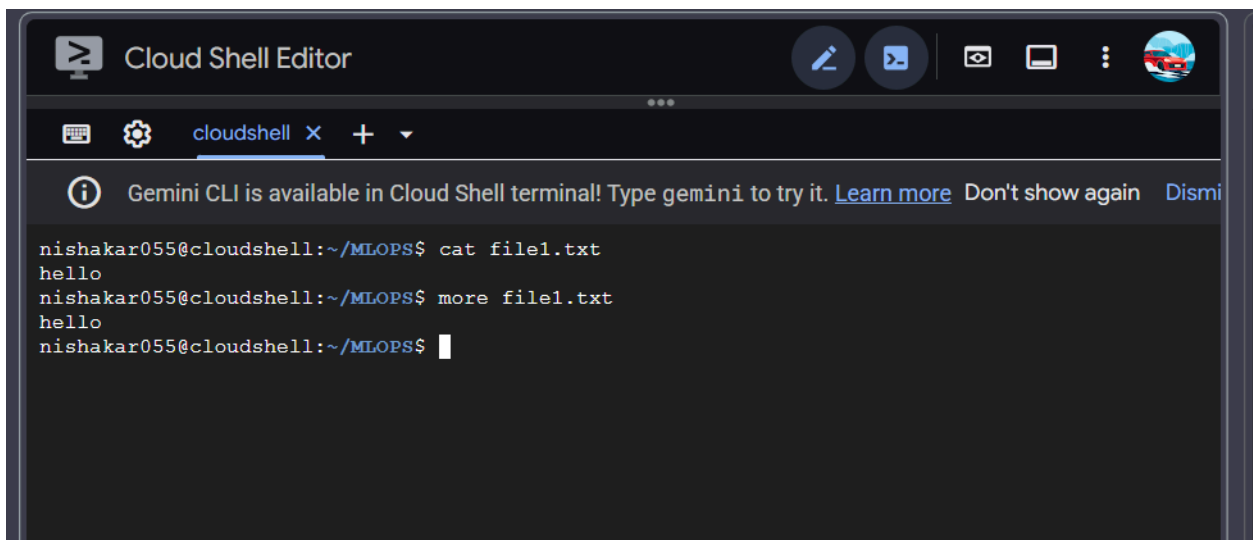
```
Cloud Shell Editor

cloudshell x + v

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't show again Dismiss

nishakar055@cloudshell:~/MLOPS/course1$ cd
nishakar055@cloudshell:~$ cd MLOPS
nishakar055@cloudshell:~/MLOPS$ ln -s file1.txt shortcut.txt
nishakar055@cloudshell:~/MLOPS$ ls
course1  shortcut.txt
nishakar055@cloudshell:~/MLOPS$ cd course1
nishakar055@cloudshell:~/MLOPS/course1$ ls
file1  MLOPS
nishakar055@cloudshell:~/MLOPS/course1$ ls -l
total 0
-rw-rw-r-- 1 nishakar055 nishakar055 0 May 14 10:20 file1
-rw-rw-r-- 1 nishakar055 nishakar055 0 May 14 10:20 MLOPS
nishakar055@cloudshell:~/MLOPS/course1$
```

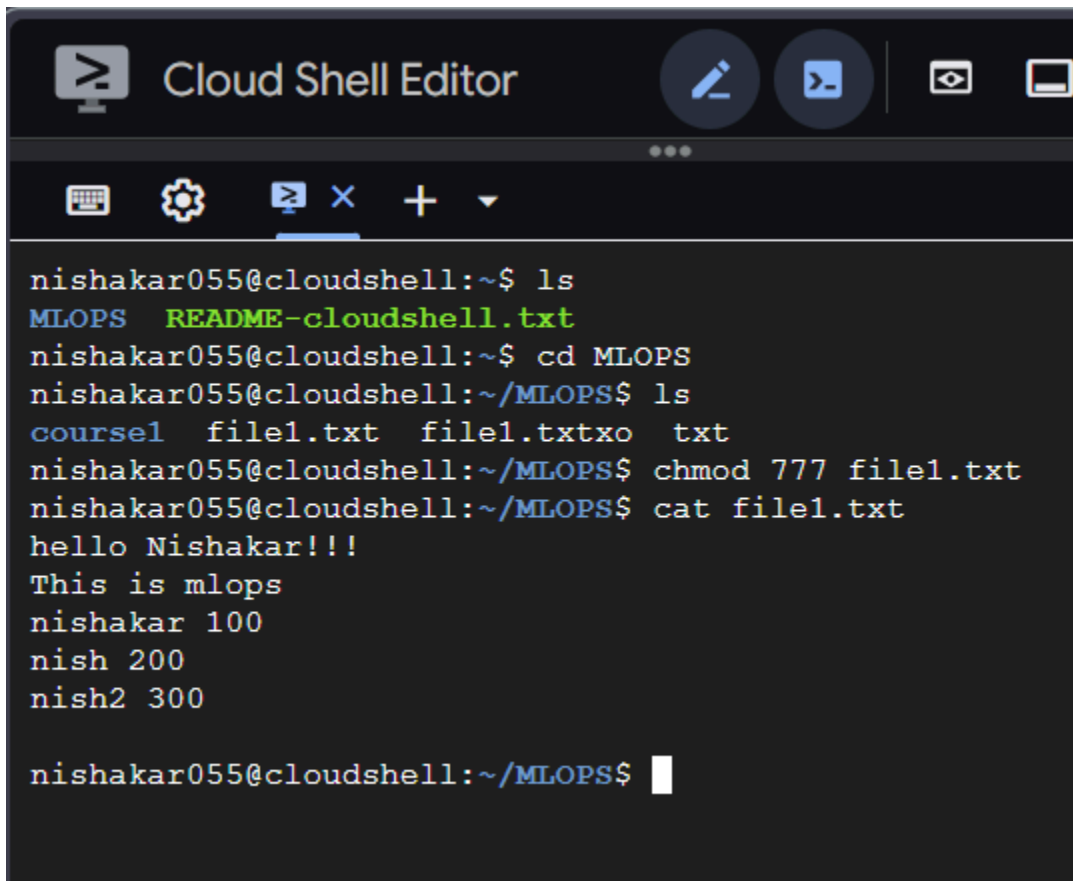
25.more



The screenshot shows the Cloud Shell Editor interface. At the top, there's a title bar with a terminal icon and the text "Cloud Shell Editor". Below it, a toolbar contains icons for keyboard shortcuts, settings, a terminal window, a plus sign, and a dropdown arrow. A notification banner at the top states: "Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#) Don't show again Dismiss". The terminal window shows the following commands and output:

```
nishakar055@cloudshell:~/MLOPS$ cat file1.txt
hello
nishakar055@cloudshell:~/MLOPS$ more file1.txt
hello
nishakar055@cloudshell:~/MLOPS$
```

26.chmod

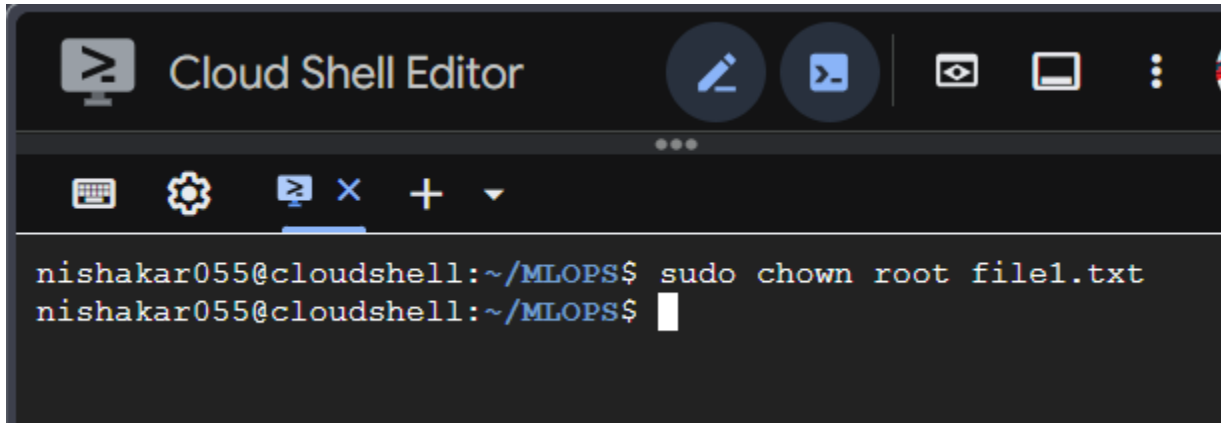


The screenshot shows the Cloud Shell Editor interface. The terminal window displays the following commands and output:

```
nishakar055@cloudshell:~$ ls
MLOPS  README-cloudshell.txt
nishakar055@cloudshell:~$ cd MLOPS
nishakar055@cloudshell:~/MLOPS$ ls
course1 file1.txt file1.txtxo txt
nishakar055@cloudshell:~/MLOPS$ chmod 777 file1.txt
nishakar055@cloudshell:~/MLOPS$ cat file1.txt
hello Nishakar!!!
This is mlops
nishakar 100
nish 200
nish2 300

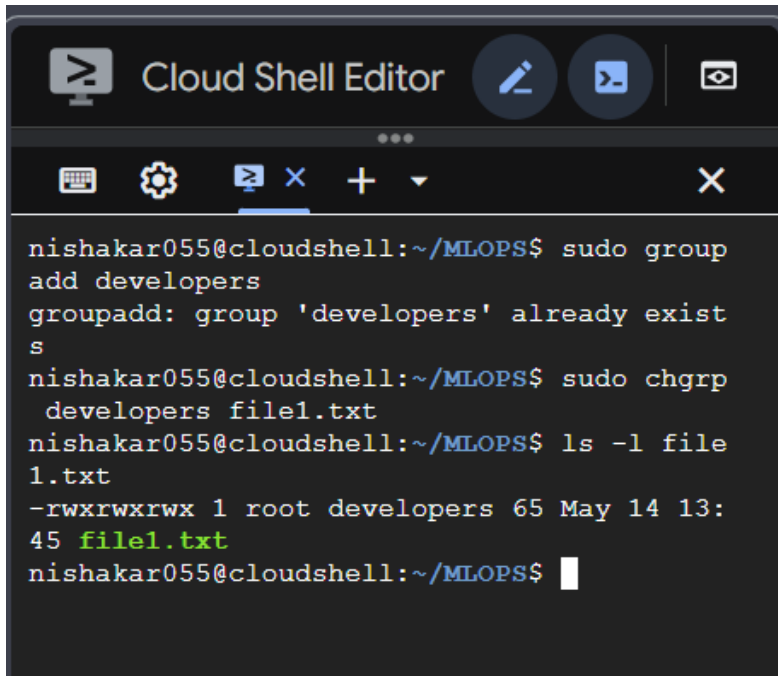
nishakar055@cloudshell:~/MLOPS$
```

27.chown



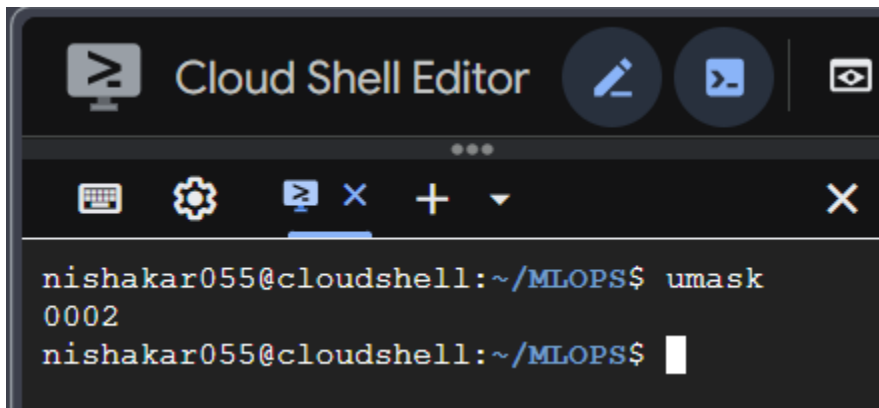
```
nishakar055@cloudshell:~/MLOPS$ sudo chown root file1.txt
nishakar055@cloudshell:~/MLOPS$
```

28.chgrp



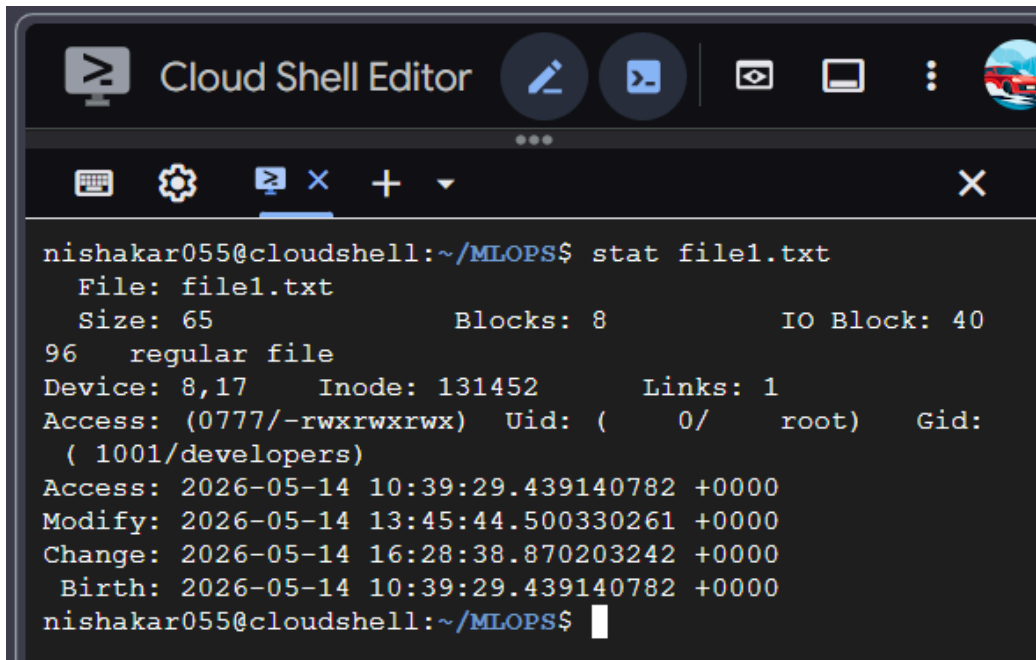
```
nishakar055@cloudshell:~/MLOPS$ sudo group
add developers
groupadd: group 'developers' already exist
s
nishakar055@cloudshell:~/MLOPS$ sudo chgrp
developers file1.txt
nishakar055@cloudshell:~/MLOPS$ ls -l file
1.txt
-rwxrwxrwx 1 root developers 65 May 14 13:
45 file1.txt
nishakar055@cloudshell:~/MLOPS$
```

29.umask



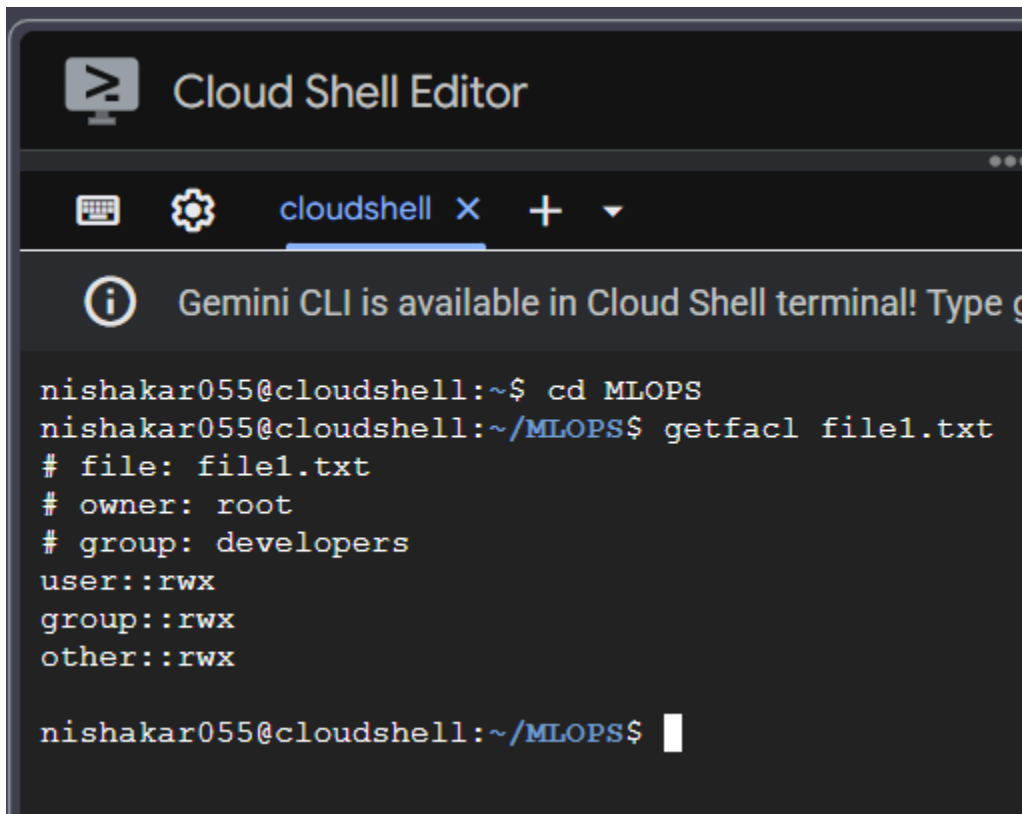
```
nishakar055@cloudshell:~/MLOPS$ umask
0002
nishakar055@cloudshell:~/MLOPS$
```

30.stat



```
nishakar055@cloudshell:~/MLOPS$ stat file1.txt
  File: file1.txt
  Size: 65          Blocks: 8          IO Block: 40
96  regular file
Device: 8,17    Inode: 131452      Links: 1
Access: (0777/-rwxrwxrwx)  Uid: (  0/   root)   Gid:
( 1001/developers)
Access: 2026-05-14 10:39:29.439140782 +0000
Modify: 2026-05-14 13:45:44.500330261 +0000
Change: 2026-05-14 16:28:38.870203242 +0000
 Birth: 2026-05-14 10:39:29.439140782 +0000
nishakar055@cloudshell:~/MLOPS$
```

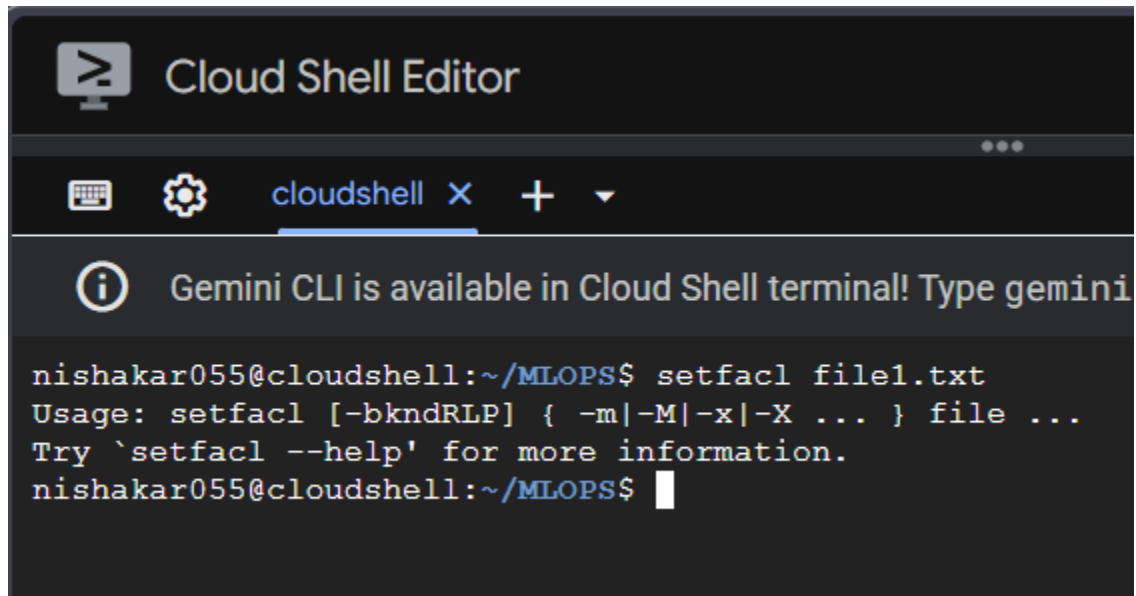
31.getfacl



```
nishakar055@cloudshell:~$ cd MLOPS
nishakar055@cloudshell:~/MLOPS$ getfacl file1.txt
# file: file1.txt
# owner: root
# group: developers
user::rwx
group::rwx
other::rwx

nishakar055@cloudshell:~/MLOPS$
```

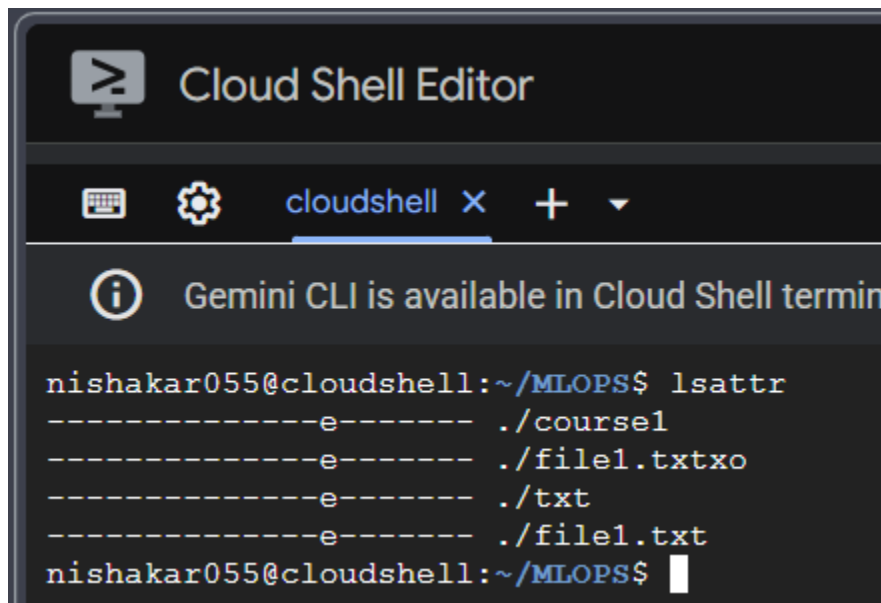
32.setfacl



The screenshot shows the Cloud Shell Editor interface. At the top, there's a title bar with a terminal icon and the text "Cloud Shell Editor". Below it is a toolbar with a keyboard icon, a settings gear, and a tab labeled "cloudshell" with a close button (X) and a plus sign (+). A notification banner below the toolbar says "Gemini CLI is available in Cloud Shell terminal! Type gemini". The terminal window shows the prompt "nishakar055@cloudshell:~/MLOPS\$". The user has entered the command "setfacl file1.txt". The terminal output shows the usage of setfacl: "Usage: setfacl [-bkndRLP] { -m|-M|-x|-X ... } file ...". It also suggests trying "setfacl --help" for more information. The prompt returns to "nishakar055@cloudshell:~/MLOPS\$".

```
nishakar055@cloudshell:~/MLOPS$ setfacl file1.txt
Usage: setfacl [-bkndRLP] { -m|-M|-x|-X ... } file ...
Try `setfacl --help' for more information.
nishakar055@cloudshell:~/MLOPS$
```

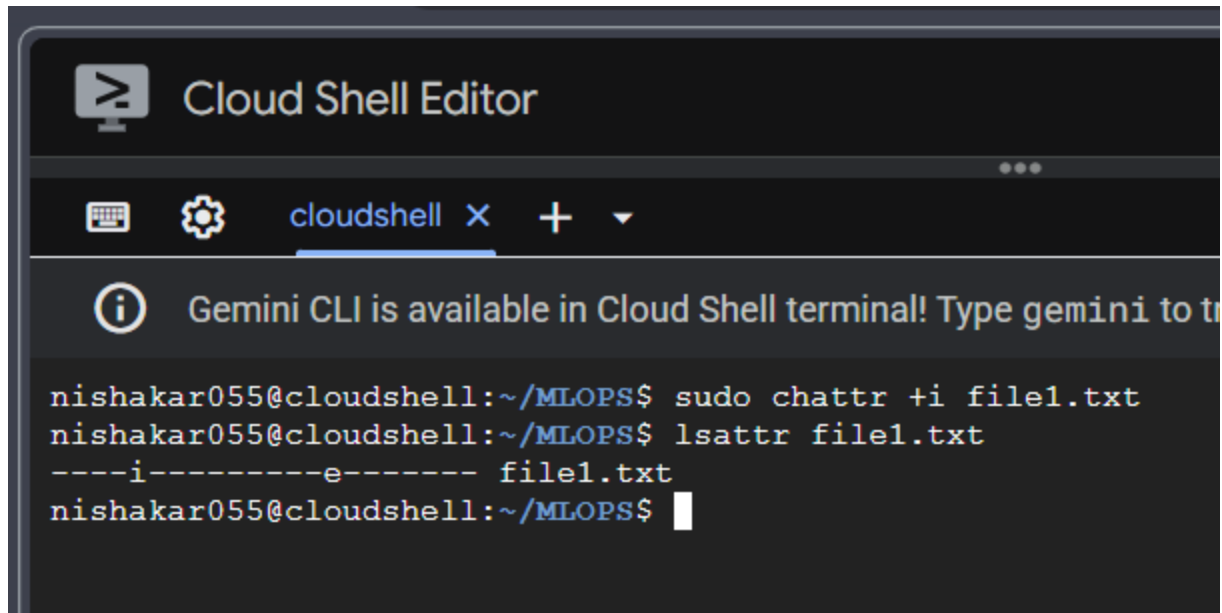
33.lsattr



The screenshot shows the Cloud Shell Editor interface. At the top, there's a title bar with a terminal icon and the text "Cloud Shell Editor". Below it is a toolbar with a keyboard icon, a settings gear, and a tab labeled "cloudshell" with a close button (X) and a plus sign (+). A notification banner below the toolbar says "Gemini CLI is available in Cloud Shell terminal! Type gemini". The terminal window shows the prompt "nishakar055@cloudshell:~/MLOPS\$". The user has entered the command "lsattr". The terminal output shows the attributes of files in the current directory: "-----e----- ./course1", "-----e----- ./file1.txt", "-----e----- ./txt", and "-----e----- ./file1.txt". The prompt returns to "nishakar055@cloudshell:~/MLOPS\$".

```
nishakar055@cloudshell:~/MLOPS$ lsattr
-----e----- ./course1
-----e----- ./file1.txt
-----e----- ./txt
-----e----- ./file1.txt
nishakar055@cloudshell:~/MLOPS$
```

34.chattr



The screenshot shows a terminal window titled "Cloud Shell Editor". The terminal prompt is `nishakar055@cloudshell:~/MLOPS$`. The user has executed the following commands:

```
sudo chattr +i file1.txt
lsattr file1.txt
```

The output of the `lsattr` command is:

```
----i-----e----- file1.txt
```

The terminal window also features a notification bar that reads: "Gemini CLI is available in Cloud Shell terminal! Type gemini to t".

35.usermod

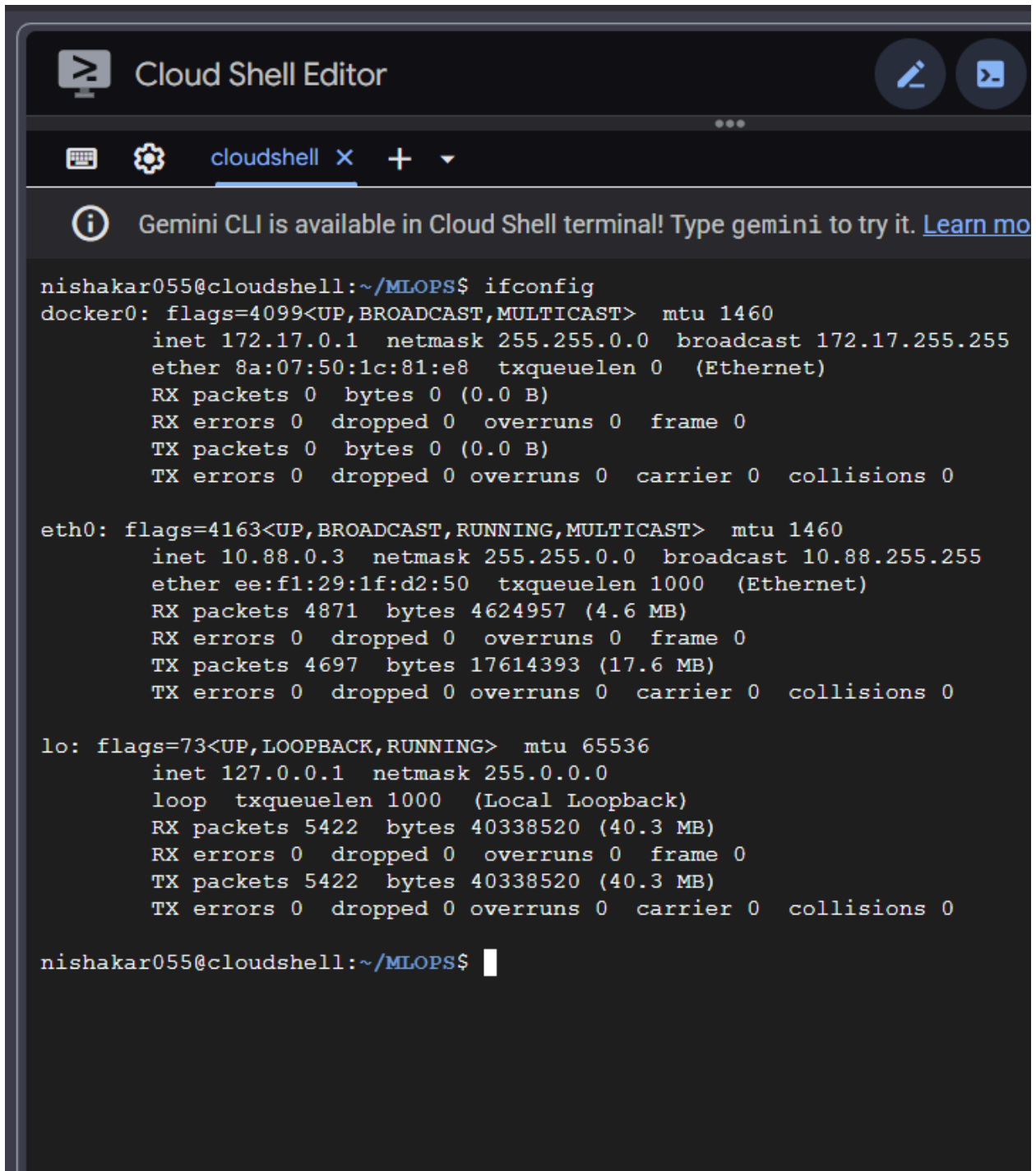
```
Cloud Shell Editor

cloudshell x + v

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't show again Dismiss

nishakar055@cloudshell:~/MLOPS$ whoami
nishakar055
nishakar055@cloudshell:~/MLOPS$ groups nishakar055
nishakar055 : nishakar055 adm sudo docker
nishakar055@cloudshell:~/MLOPS$ sudo usermod -U nishakar055
usermod: unlocking the user's password would result in a passwordless account.
You should set a password with usermod -p to unlock this user's password.
nishakar055@cloudshell:~/MLOPS$ sudo usermod -aG sudo nishakar055
nishakar055@cloudshell:~/MLOPS$ id nishakar055
uid=1000(nishakar055) gid=1000(nishakar055) groups=1000(nishakar055),4(adm),27(sudo),996(docke
nishakar055@cloudshell:~/MLOPS$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
messagebus:x:100:101::/nonexistent:/usr/sbin/nologin
polkitd:x:997:997:User for polkitd:/:/usr/sbin/nologin
syslog:x:101:102::/nonexistent:/usr/sbin/nologin
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
dhcpcd:x:102:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
redis:x:103:104::/var/lib/redis:/usr/sbin/nologin
sshd:x:104:65534::/run/sshd:/usr/sbin/nologin
postgres:x:105:106:PostgreSQL administrator,,,:/var/lib/postgresql:/bin/bash
nishakar055:x:1000:1000:~/home/nishakar055:/bin/bash
nishakar055@cloudshell:~/MLOPS$
```

36.ifconfig



The image shows a Cloud Shell Editor interface. At the top, there's a header bar with a terminal icon, the text "Cloud Shell Editor", and two circular icons on the right (a pencil and a document). Below the header is a tab bar with a keyboard icon, a gear icon, and a tab labeled "cloudshell" with a close button (X) and a plus sign (+). A notification banner below the tabs says "Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#)". The main terminal area has a dark background with light-colored text. The prompt is "nishakar055@cloudshell:~/MLOPS\$". The command "ifconfig" has been executed, showing details for three network interfaces: "docker0", "eth0", and "lo". Each interface listing includes flags, mtu, inet address, netmask, broadcast address, ether address, txqueuelen, and statistics for RX and TX packets, bytes, errors, dropped, overruns, carrier, and collisions. The "lo" interface is the loopback interface. The prompt "nishakar055@cloudshell:~/MLOPS\$" is shown again at the bottom with a cursor.

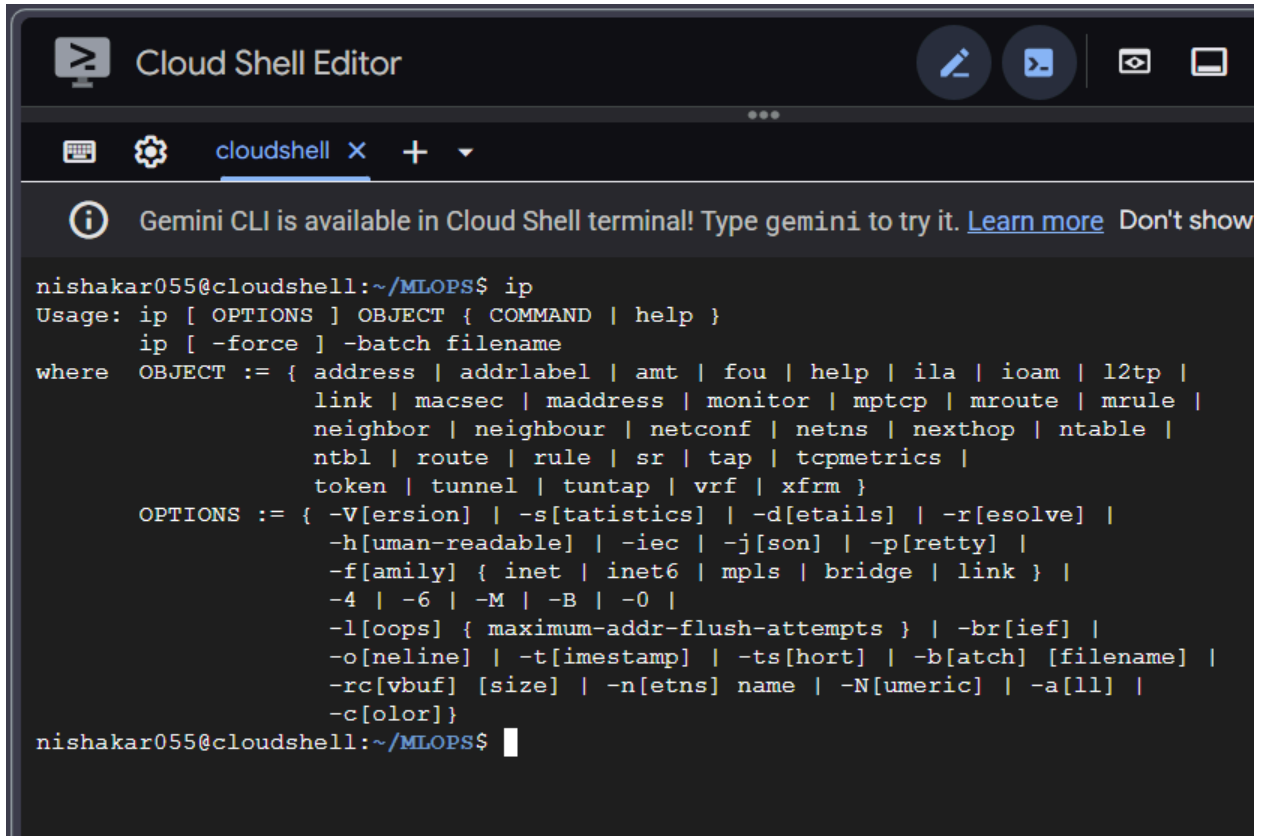
```
nishakar055@cloudshell:~/MLOPS$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1460
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 8a:07:50:1c:81:e8 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1460
    inet 10.88.0.3 netmask 255.255.0.0 broadcast 10.88.255.255
    ether ee:f1:29:1f:d2:50 txqueuelen 1000 (Ethernet)
    RX packets 4871 bytes 4624957 (4.6 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 4697 bytes 17614393 (17.6 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 5422 bytes 40338520 (40.3 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 5422 bytes 40338520 (40.3 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

nishakar055@cloudshell:~/MLOPS$
```

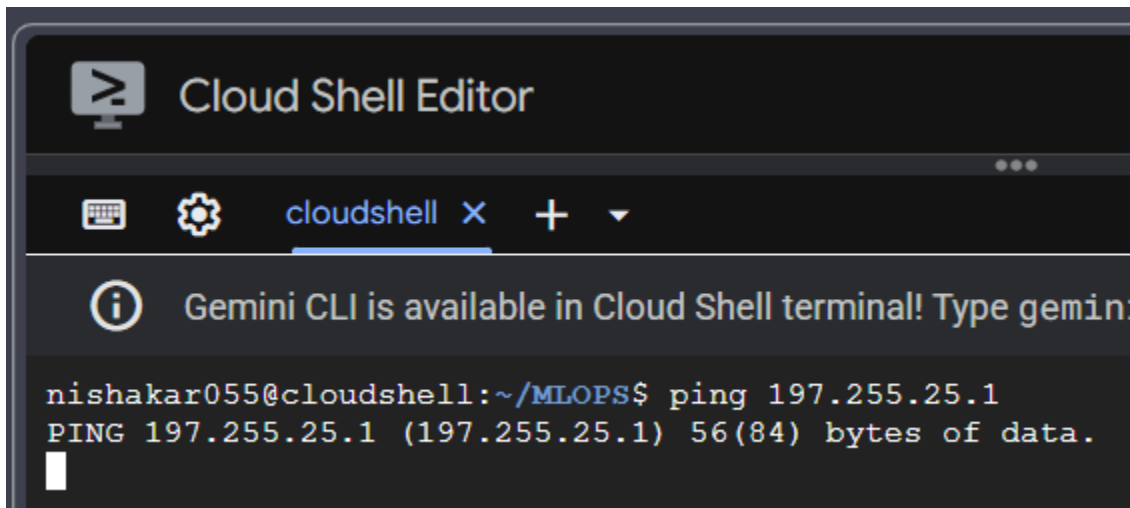
37.ip



The image shows a Cloud Shell Editor terminal window. The title bar says "Cloud Shell Editor". Below the title bar, there's a tab labeled "cloudshell" with a close button and a plus sign. A notification banner at the top says "Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#) Don't show". The terminal content shows the user running the command `ip` in a shell prompt `nishakar055@cloudshell:~/MLOPS$`. The output is the usage and options for the `ip` command.

```
nishakar055@cloudshell:~/MLOPS$ ip
Usage: ip [ OPTIONS ] OBJECT { COMMAND | help }
       ip [ -force ] -batch filename
where  OBJECT := { address | addrlabel | amt | fou | help | ila | ioam | l2tp |
                  link | macsec | maddress | monitor | mptcp | mroute | mrule |
                  neighbor | neighbour | netconf | netns | nexthop | ntable |
                  ntbl | route | rule | sr | tap | tcpmetrics |
                  token | tunnel | tuntap | vrf | xfrm }
       OPTIONS := { -V[ersion] | -s[tatistics] | -d[etails] | -r[esolve] |
                   -h[uman-readable] | -iec | -j[son] | -p[retty] |
                   -f[amily] { inet | inet6 | mpls | bridge | link } |
                   -4 | -6 | -M | -B | -O |
                   -l[oops] { maximum-addr-flush-attempts } | -br[ief] |
                   -o[neline] | -t[imestamp] | -ts[hort] | -b[atch] [filename] |
                   -rc[vbuf] [size] | -n[etns] name | -N[umeric] | -a[ll] |
                   -c[olor]}
```

38.ping



The image shows a Cloud Shell Editor terminal window. The title bar says "Cloud Shell Editor". Below the title bar, there's a tab labeled "cloudshell" with a close button and a plus sign. A notification banner at the top says "Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#) Don't show". The terminal content shows the user running the command `ping 197.255.25.1` in a shell prompt `nishakar055@cloudshell:~/MLOPS$`. The output is the standard ping output.

```
nishakar055@cloudshell:~/MLOPS$ ping 197.255.25.1
PING 197.255.25.1 (197.255.25.1) 56(84) bytes of data.
```

39.netstat



Cloud Shell Editor



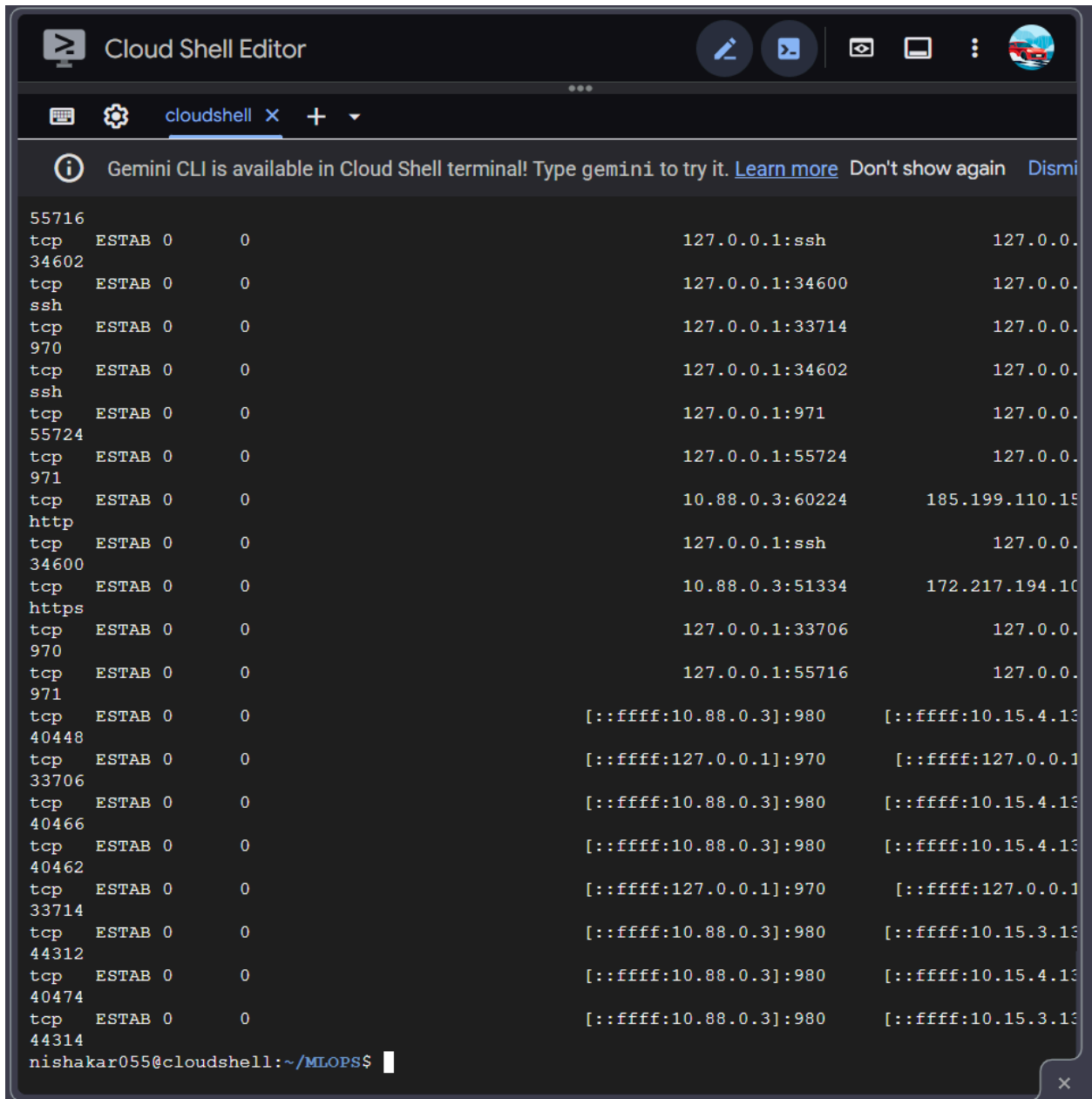
cloudshell X + ▾



Gemini CLI is available in Cloud Shell terminal! Type `gemini` to try it. [Learn more](#)

```
unix 3      [ ]      STREAM    CONNECTED 68263
unix 3      [ ]      STREAM    CONNECTED 71151
unix 3      [ ]      STREAM    CONNECTED 71149
unix 3      [ ]      STREAM    CONNECTED 71148
unix 3      [ ]      STREAM    CONNECTED 68542
unix 3      [ ]      STREAM    CONNECTED 68538
unix 3      [ ]      STREAM    CONNECTED 69525
unix 3      [ ]      STREAM    CONNECTED 69523
unix 2      [ ]      STREAM    CONNECTED 69137
unix 3      [ ]      STREAM    CONNECTED 68541
unix 3      [ ]      STREAM    CONNECTED 68264
unix 3      [ ]      STREAM    CONNECTED 69504
unix 2      [ ]      DGRAM     CONNECTED 68766
unix 3      [ ]      STREAM    CONNECTED 69528
unix 3      [ ]      STREAM    CONNECTED 69069
unix 2      [ ]      DGRAM     CONNECTED 69038
unix 3      [ ]      STREAM    CONNECTED 69510
unix 2      [ ]      DGRAM     CONNECTED 67495
unix 2      [ ]      DGRAM     CONNECTED 68178
unix 3      [ ]      STREAM    CONNECTED 68535
unix 3      [ ]      STREAM    CONNECTED 69179
unix 3      [ ]      STREAM    CONNECTED 69526
unix 9      [ ]      DGRAM     CONNECTED 67908    /dev/log
unix 3      [ ]      STREAM    CONNECTED 68540
unix 3      [ ]      STREAM    CONNECTED 69413
unix 3      [ ]      STREAM    CONNECTED 69524
unix 3      [ ]      STREAM    CONNECTED 69511
unix 2      [ ]      DGRAM     CONNECTED 68258
unix 3      [ ]      STREAM    CONNECTED 67501
unix 3      [ ]      STREAM    CONNECTED 69658
unix 3      [ ]      STREAM    CONNECTED 69506
unix 3      [ ]      STREAM    CONNECTED 69501
unix 3      [ ]      STREAM    CONNECTED 71153
unix 3      [ ]      STREAM    CONNECTED 69508
unix 2      [ ]      STREAM    CONNECTED 69104
unix 3      [ ]      STREAM    CONNECTED 66339
unix 3      [ ]      STREAM    CONNECTED 71152
unix 3      [ ]      STREAM    CONNECTED 71150
unix 3      [ ]      STREAM    CONNECTED 69502
nishakar055@cloudshell:~/MLOPS$
```

40.ss



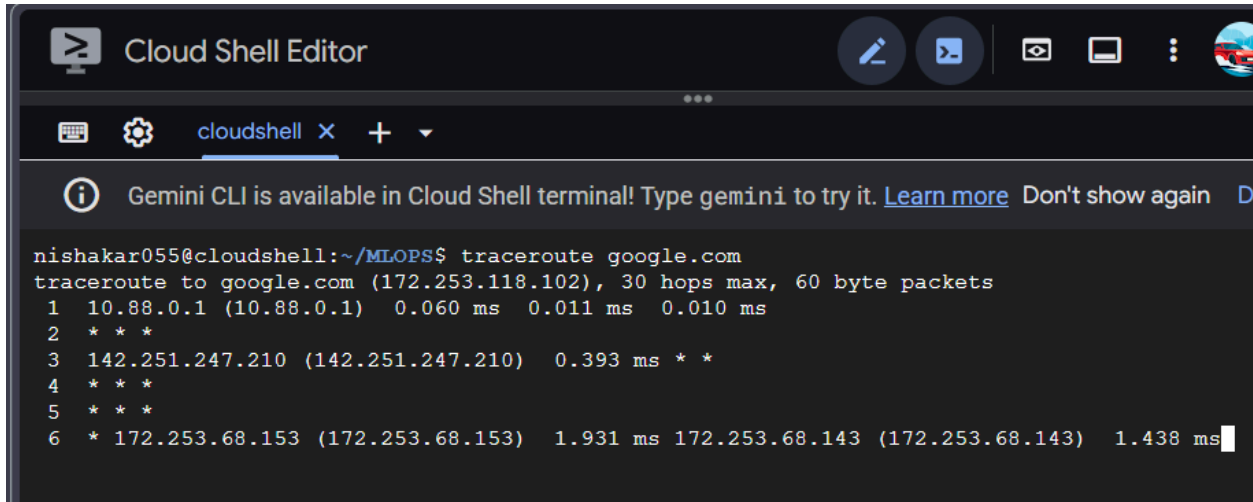
```
Cloud Shell Editor

cloudshell x + v

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't show again Dismiss

55716
tcp ESTAB 0 0 127.0.0.1:ssh 127.0.0.1:55716
34602
tcp ESTAB 0 0 127.0.0.1:34600 127.0.0.1:34602
ssh
tcp ESTAB 0 0 127.0.0.1:33714 127.0.0.1:33714
970
tcp ESTAB 0 0 127.0.0.1:34602 127.0.0.1:34602
ssh
tcp ESTAB 0 0 127.0.0.1:971 127.0.0.1:971
55724
tcp ESTAB 0 0 127.0.0.1:55724 127.0.0.1:55724
971
tcp ESTAB 0 0 10.88.0.3:60224 185.199.110.15:60224
http
tcp ESTAB 0 0 127.0.0.1:ssh 127.0.0.1:55716
34600
tcp ESTAB 0 0 10.88.0.3:51334 172.217.194.10:51334
https
tcp ESTAB 0 0 127.0.0.1:33706 127.0.0.1:33706
970
tcp ESTAB 0 0 127.0.0.1:55716 127.0.0.1:55716
971
tcp ESTAB 0 0 ::ffff:10.88.0.3:980 ::ffff:10.15.4.13:980
40448
tcp ESTAB 0 0 ::ffff:127.0.0.1:970 ::ffff:127.0.0.1:970
33706
tcp ESTAB 0 0 ::ffff:10.88.0.3:980 ::ffff:10.15.4.13:980
40466
tcp ESTAB 0 0 ::ffff:10.88.0.3:980 ::ffff:10.15.4.13:980
40462
tcp ESTAB 0 0 ::ffff:127.0.0.1:970 ::ffff:127.0.0.1:970
33714
tcp ESTAB 0 0 ::ffff:10.88.0.3:980 ::ffff:10.15.3.13:980
44312
tcp ESTAB 0 0 ::ffff:10.88.0.3:980 ::ffff:10.15.4.13:980
40474
tcp ESTAB 0 0 ::ffff:10.88.0.3:980 ::ffff:10.15.3.13:980
44314
nishakar055@cloudshell:~/MLOPS$
```

41.traceroute



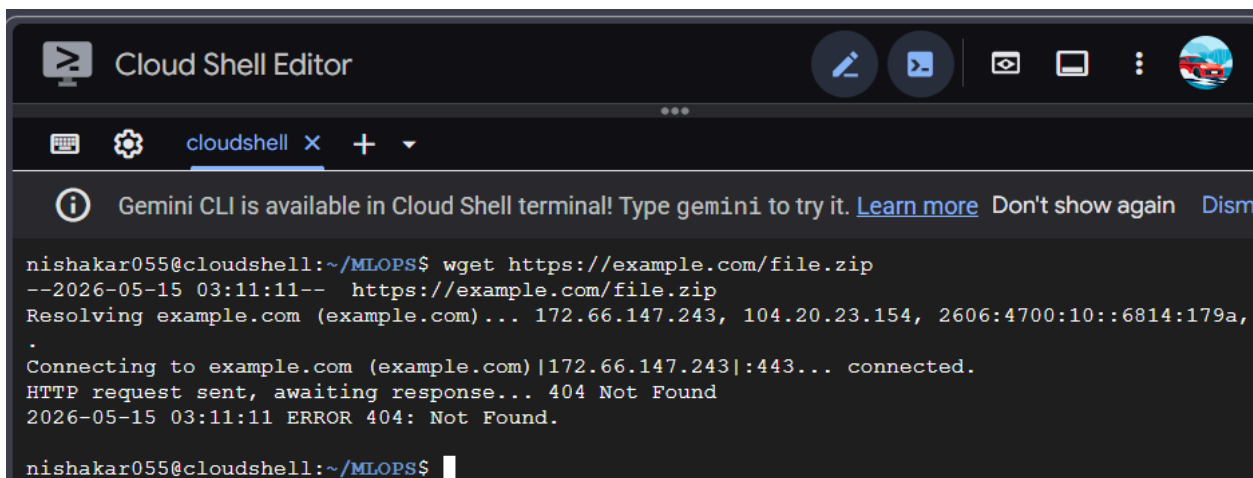
```
Cloud Shell Editor

cloudshell x + v

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't show again D

nishakar055@cloudshell:~/MLOPS$ traceroute google.com
traceroute to google.com (172.253.118.102), 30 hops max, 60 byte packets
 1  10.88.0.1 (10.88.0.1)  0.060 ms  0.011 ms  0.010 ms
 2  * * *
 3  142.251.247.210 (142.251.247.210)  0.393 ms  * *
 4  * * *
 5  * * *
 6  * 172.253.68.153 (172.253.68.153)  1.931 ms 172.253.68.143 (172.253.68.143)  1.438 ms
```

42.wget



```
Cloud Shell Editor

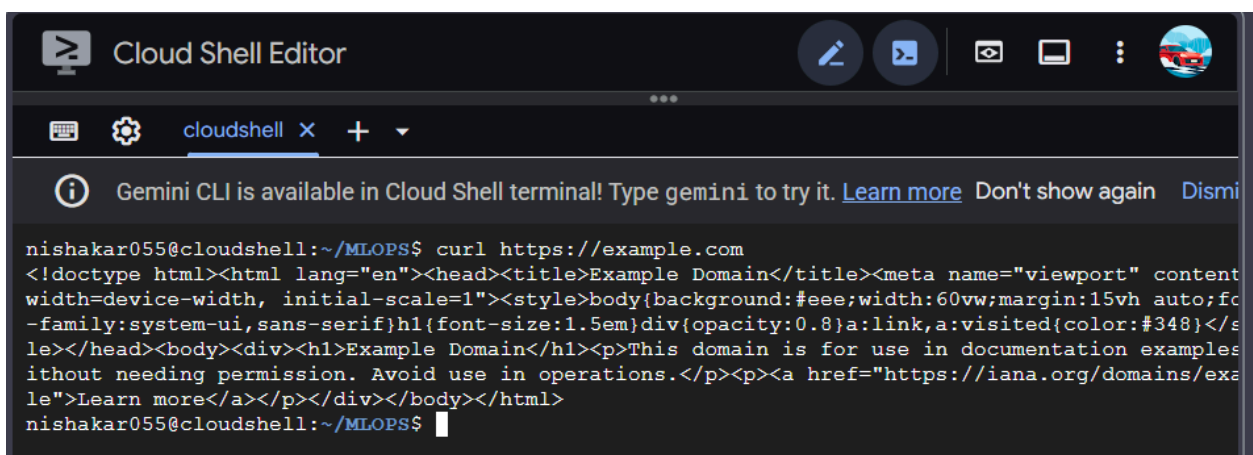
cloudshell x + v

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't show again Dismi

nishakar055@cloudshell:~/MLOPS$ wget https://example.com/file.zip
--2026-05-15 03:11:11-- https://example.com/file.zip
Resolving example.com (example.com)... 172.66.147.243, 104.20.23.154, 2606:4700:10::6814:179a,
.
Connecting to example.com (example.com)|172.66.147.243|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-05-15 03:11:11 ERROR 404: Not Found.

nishakar055@cloudshell:~/MLOPS$
```

43.curl



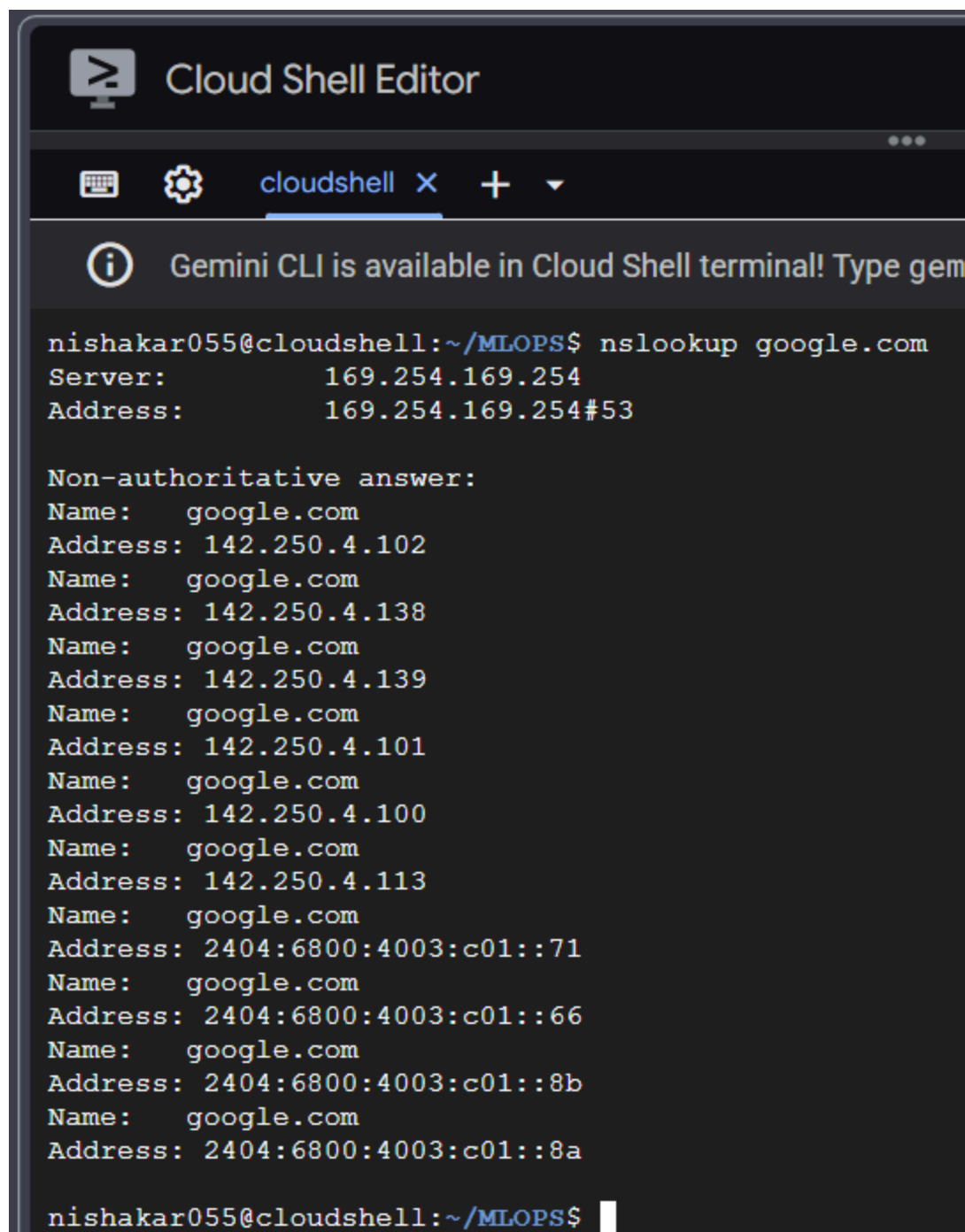
```
Cloud Shell Editor

cloudshell x + v

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't show again Dismi

nishakar055@cloudshell:~/MLOPS$ curl https://example.com
<!doctype html><html lang="en"><head><title>Example Domain</title><meta name="viewport" content
width=device-width, initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;fc
-family:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</s
le></head><body><div><h1>Example Domain</h1><p>This domain is for use in documentation examples
ithout needing permission. Avoid use in operations.</p><p><a href="https://iana.org/domains/exa
le">Learn more</a></p></div></body></html>
nishakar055@cloudshell:~/MLOPS$
```

44. Nslookup



The image shows a Cloud Shell Editor window. At the top, there's a title bar with a terminal icon and the text "Cloud Shell Editor". Below the title bar is a toolbar with a keyboard icon, a gear icon, and a tab labeled "cloudshell" with a close button and a plus sign. A notification banner at the top of the terminal area says "Gemini CLI is available in Cloud Shell terminal! Type gem". The terminal content shows a user running the command `nslookup google.com`. The output displays the server and address for the first lookup, followed by a "Non-authoritative answer:" section listing multiple IP addresses for google.com. The prompt at the bottom is `nishakar055@cloudshell:~/MLOPS$`.

```
nishakar055@cloudshell:~/MLOPS$ nslookup google.com
Server:          169.254.169.254
Address:         169.254.169.254#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.4.102
Name:   google.com
Address: 142.250.4.138
Name:   google.com
Address: 142.250.4.139
Name:   google.com
Address: 142.250.4.101
Name:   google.com
Address: 142.250.4.100
Name:   google.com
Address: 142.250.4.113
Name:   google.com
Address: 2404:6800:4003:c01::71
Name:   google.com
Address: 2404:6800:4003:c01::66
Name:   google.com
Address: 2404:6800:4003:c01::8b
Name:   google.com
Address: 2404:6800:4003:c01::8a

nishakar055@cloudshell:~/MLOPS$
```

45. Dig


```
Cloud Shell Editor

cloudshell x + v

(i) Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more

nishakar055@cloudshell:~/MLOPS$ dig

; <<>> DiG 9.18.39-0ubuntu0.24.04.3-Ubuntu <<>>
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 23569
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 13, AUTHORITY: 0, ADDITIONAL: 1

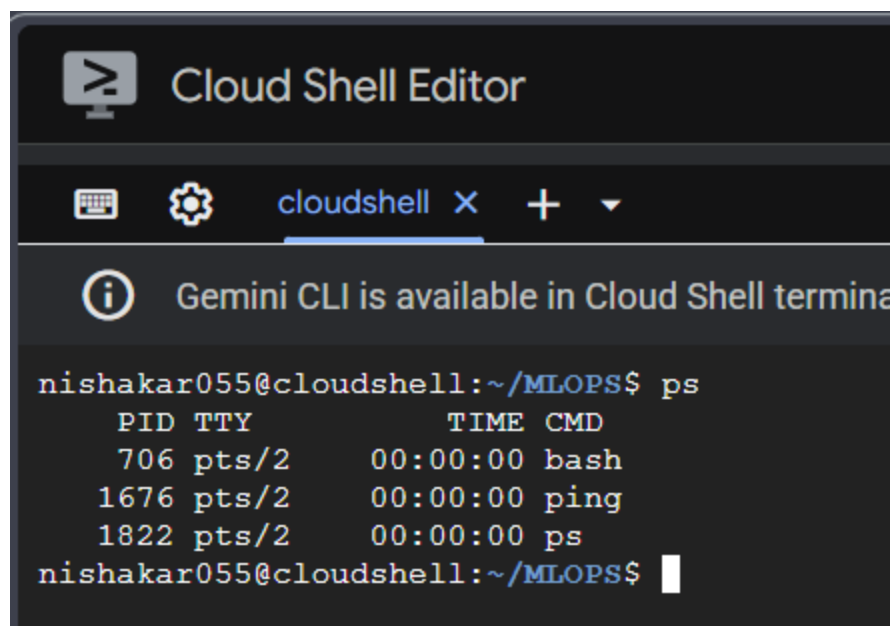
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;.                               IN      NS

;; ANSWER SECTION:
.           259200  IN      NS      a.root-servers.net.
.           259200  IN      NS      b.root-servers.net.
.           259200  IN      NS      c.root-servers.net.
.           259200  IN      NS      d.root-servers.net.
.           259200  IN      NS      e.root-servers.net.
.           259200  IN      NS      f.root-servers.net.
.           259200  IN      NS      g.root-servers.net.
.           259200  IN      NS      h.root-servers.net.
.           259200  IN      NS      i.root-servers.net.
.           259200  IN      NS      j.root-servers.net.
.           259200  IN      NS      k.root-servers.net.
.           259200  IN      NS      l.root-servers.net.
.           259200  IN      NS      m.root-servers.net.

;; Query time: 3 msec
;; SERVER: 169.254.169.254#53(169.254.169.254) (UDP)
;; WHEN: Fri May 15 03:13:08 UTC 2026
;; MSG SIZE rcvd: 239

nishakar055@cloudshell:~/MLOPS$
```

46.ps



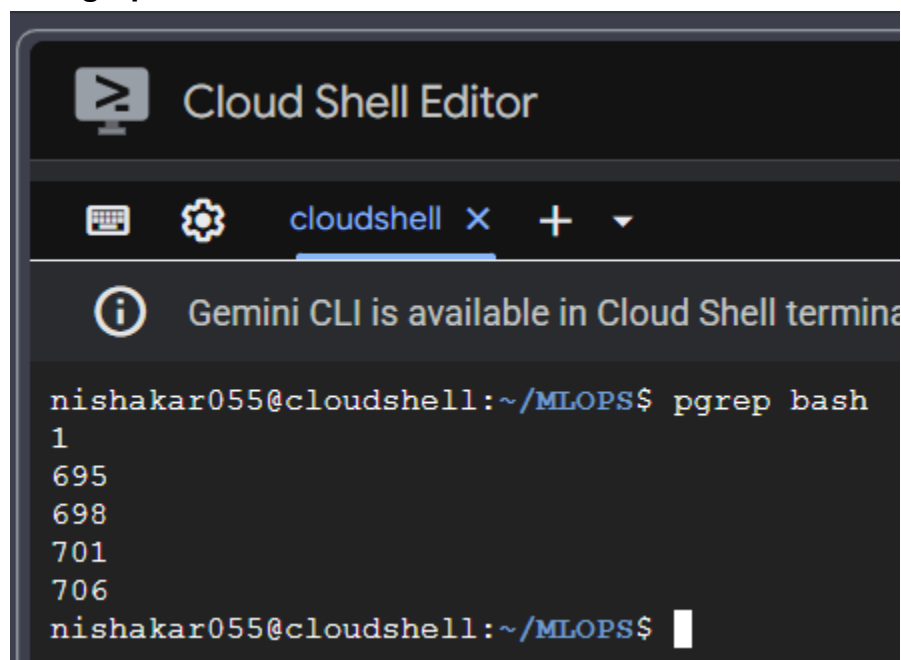
The image shows a Cloud Shell Editor terminal window. The title bar says "Cloud Shell Editor". Below the title bar, there is a toolbar with a keyboard icon, a gear icon, and a tab labeled "cloudshell" with a close button (X) and a plus sign (+). A notification banner at the top of the terminal area says "Gemini CLI is available in Cloud Shell terminal". The terminal content shows the user "nishakar055" at the prompt "cloudshell:~/MLOPS\$". They have entered the command "ps". The output is a table with four columns: PID, TTY, TIME, and CMD. The table lists three processes: a bash shell (PID 706), a ping command (PID 1676), and the ps command itself (PID 1822).

```
nishakar055@cloudshell:~/MLOPS$ ps
```

PID	TTY	TIME	CMD
706	pts/2	00:00:00	bash
1676	pts/2	00:00:00	ping
1822	pts/2	00:00:00	ps

```
nishakar055@cloudshell:~/MLOPS$
```

47. Pgrep

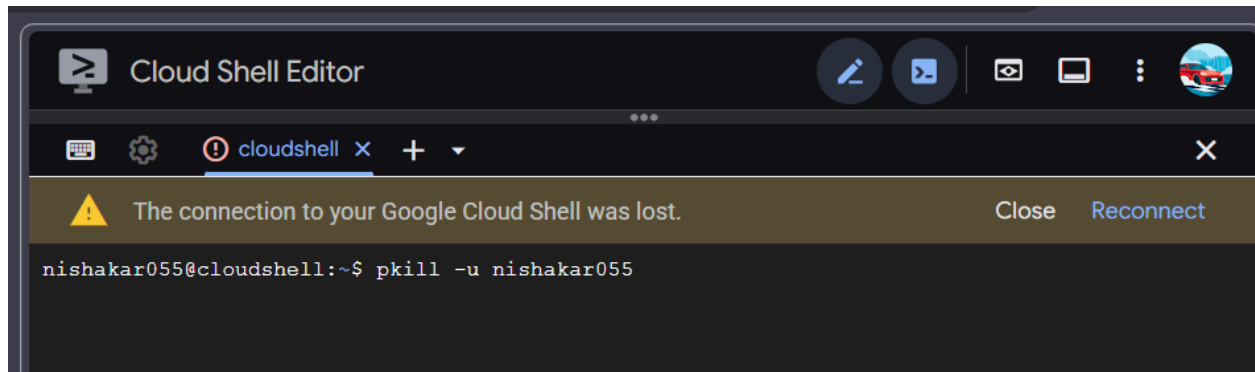


The image shows a Cloud Shell Editor terminal window, similar to the one above. The title bar says "Cloud Shell Editor". The toolbar and notification banner are the same. The terminal content shows the user "nishakar055" at the prompt "cloudshell:~/MLOPS\$". They have entered the command "pgrep bash". The output is a list of PIDs: 1, 695, 698, 701, and 706.

```
nishakar055@cloudshell:~/MLOPS$ pgrep bash
```

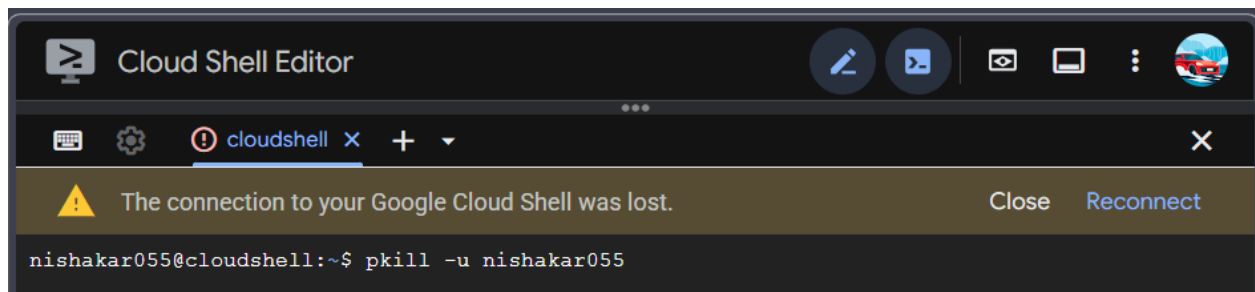
```
1
695
698
701
706
nishakar055@cloudshell:~/MLOPS$
```

48. Pkill



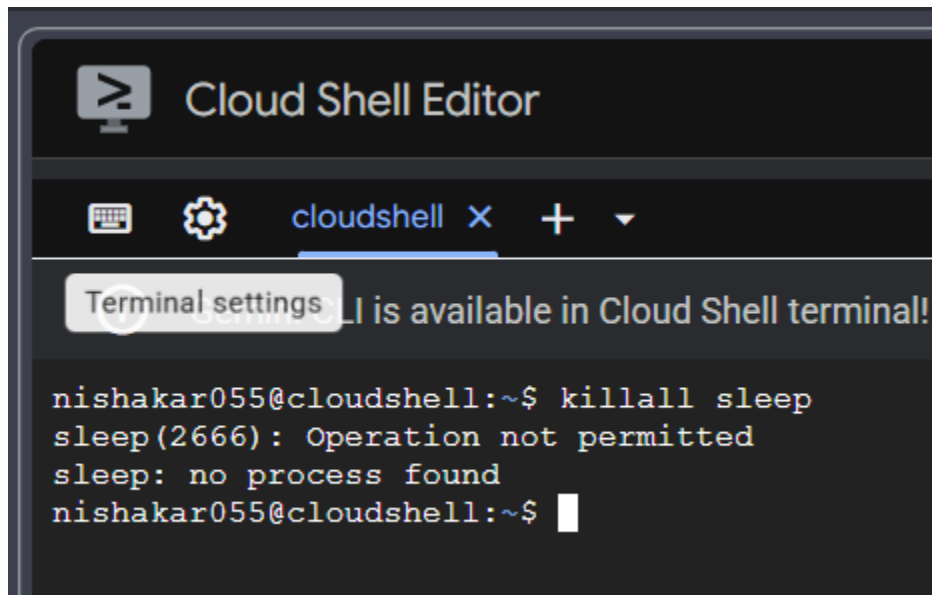
The screenshot shows the Cloud Shell Editor interface. At the top, there's a title bar with a terminal icon and the text "Cloud Shell Editor". Below it, a tab bar shows a "cloudshell" tab with a close button. A yellow warning banner states "The connection to your Google Cloud Shell was lost." with "Close" and "Reconnect" buttons. The terminal window displays the command `nishakar055@cloudshell:~$ pkill -u nishakar055`.

49. Kill



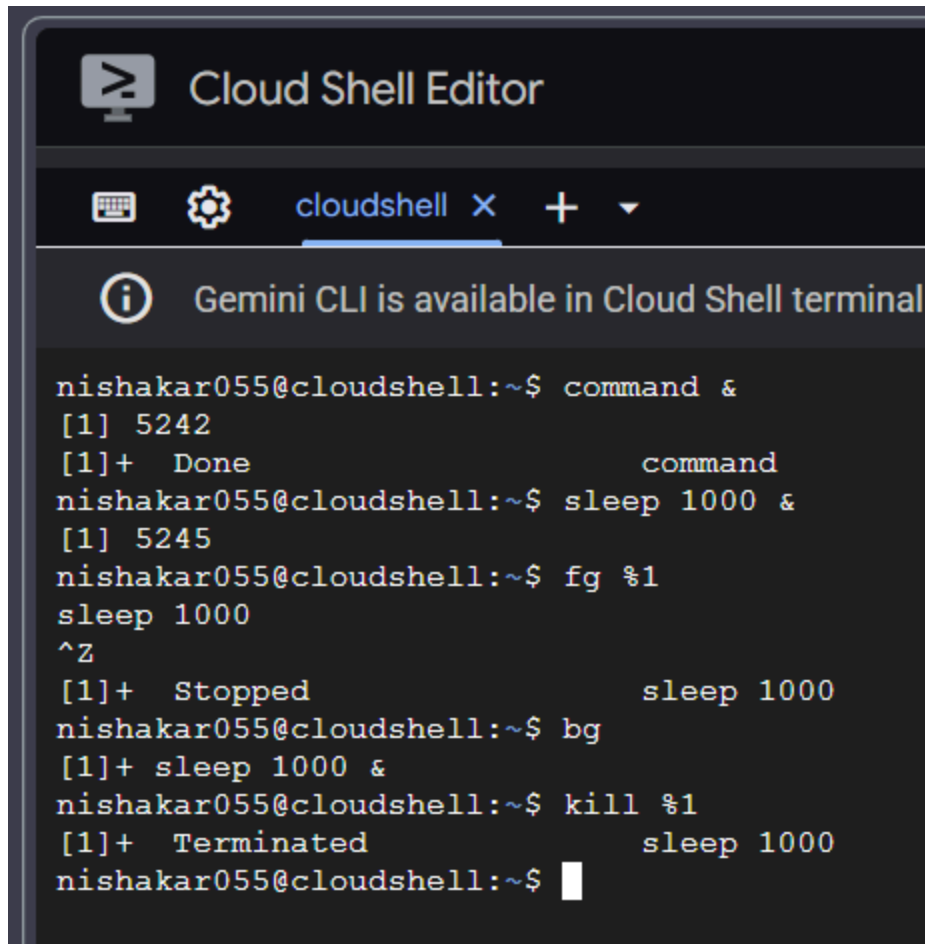
The screenshot shows the Cloud Shell Editor interface. At the top, there's a title bar with a terminal icon and the text "Cloud Shell Editor". Below it, a tab bar shows a "cloudshell" tab with a close button. A yellow warning banner states "The connection to your Google Cloud Shell was lost." with "Close" and "Reconnect" buttons. The terminal window displays the command `nishakar055@cloudshell:~$ kill -u nishakar055`.

50. Killall



The screenshot shows the Cloud Shell Editor interface. At the top, there's a title bar with a terminal icon and the text "Cloud Shell Editor". Below it, a tab bar shows a "cloudshell" tab with a close button. A tooltip labeled "Terminal settings" is visible over the terminal window. The terminal window displays the command `nishakar055@cloudshell:~$ killall sleep` and its output: `sleep(2666): Operation not permitted` and `sleep: no process found`. The prompt `nishakar055@cloudshell:~$` is shown at the bottom.

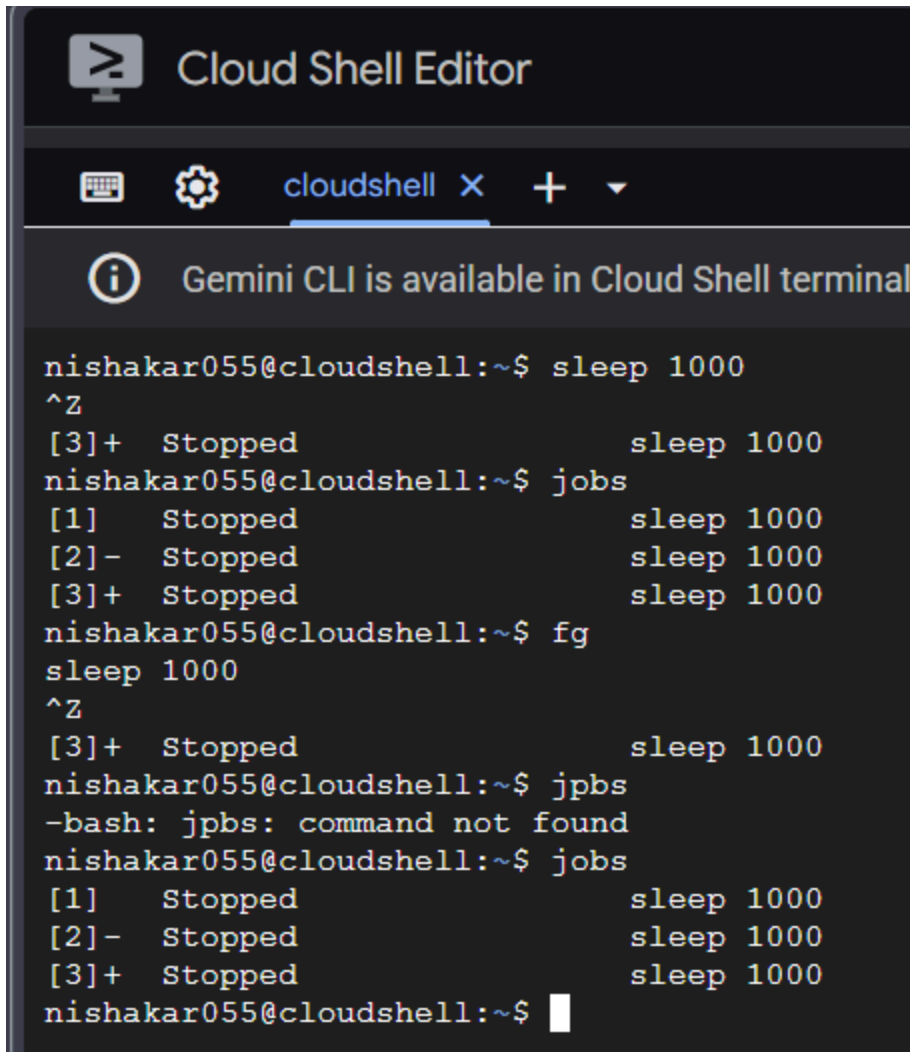
51. Bg



The screenshot shows the Cloud Shell Editor interface. At the top, there's a title bar with a terminal icon and the text "Cloud Shell Editor". Below it is a toolbar with a keyboard icon, a gear icon, and a tab labeled "cloudshell" with a close button (X) and a plus sign (+). A notification banner below the toolbar says "Gemini CLI is available in Cloud Shell terminal!". The main area is a terminal window with the following text:

```
nishakar055@cloudshell:~$ command &
[1] 5242
[1]+  Done                  command
nishakar055@cloudshell:~$ sleep 1000 &
[1] 5245
nishakar055@cloudshell:~$ fg %1
sleep 1000
^Z
[1]+  Stopped                  sleep 1000
nishakar055@cloudshell:~$ bg
[1]+ sleep 1000 &
nishakar055@cloudshell:~$ kill %1
[1]+  Terminated            sleep 1000
nishakar055@cloudshell:~$
```

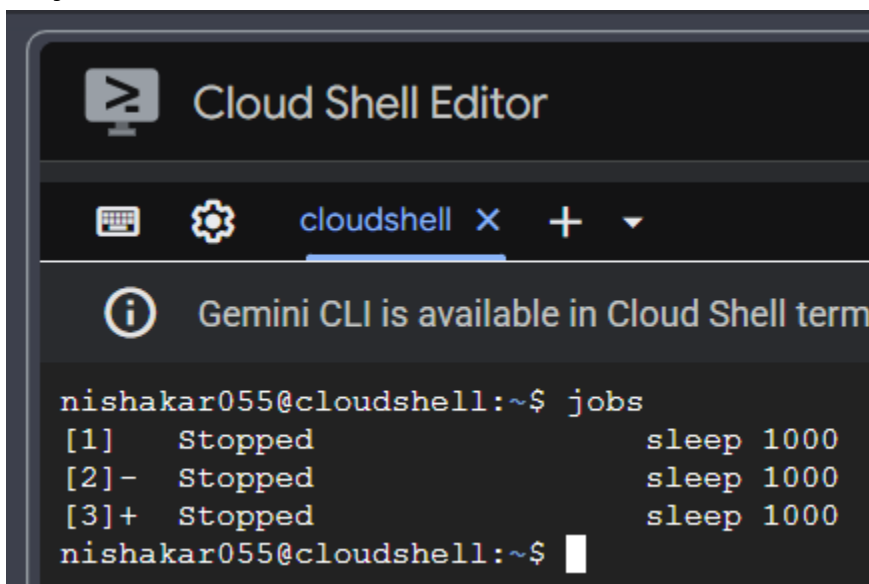
52. Fg



The image shows a Cloud Shell Editor terminal window. At the top, there's a title bar with a terminal icon and the text "Cloud Shell Editor". Below it is a toolbar with a keyboard icon, a gear icon, and a tab labeled "cloudshell" with a close button (X) and a plus sign (+). A notification bar below the toolbar says "Gemini CLI is available in Cloud Shell terminal". The terminal content shows a user named "nishakar055" at "cloudshell" in the home directory (~). They run "sleep 1000", press Ctrl-Z (^Z), and see "[3]+ Stopped sleep 1000". Then they run "jobs" and see a list of three stopped jobs. They press Ctrl-Z again, and the jobs are shown as running. Then they run "fg", which brings the first job back to the foreground. They press Ctrl-Z again, and the job is shown as stopped. Finally, they run "jobs" and see the same list of three stopped jobs.

```
nishakar055@cloudshell:~$ sleep 1000
^Z
[3]+  Stopped                  sleep 1000
nishakar055@cloudshell:~$ jobs
[1]  Stopped                  sleep 1000
[2]-  Stopped                  sleep 1000
[3]+  Stopped                  sleep 1000
nishakar055@cloudshell:~$ fg
sleep 1000
^Z
[3]+  Stopped                  sleep 1000
nishakar055@cloudshell:~$ jpbs
-bash: jpbs: command not found
nishakar055@cloudshell:~$ jobs
[1]  Stopped                  sleep 1000
[2]-  Stopped                  sleep 1000
[3]+  Stopped                  sleep 1000
nishakar055@cloudshell:~$
```

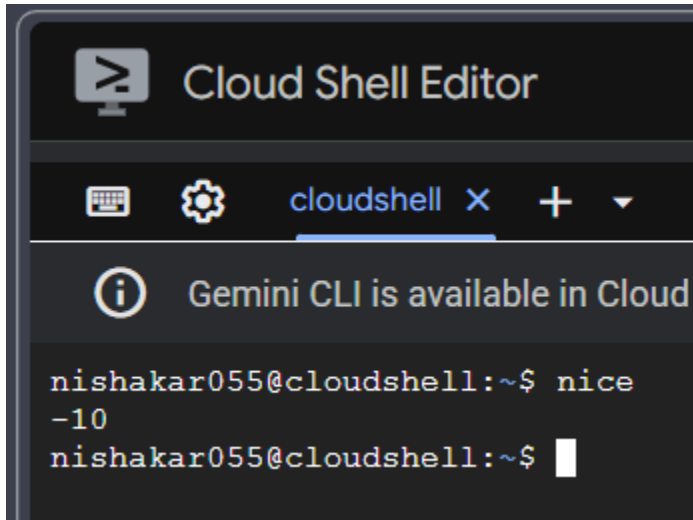
53. jobs



The image shows a Cloud Shell Editor terminal window, similar to the one above. It shows the output of the "jobs" command. The terminal content shows a user named "nishakar055" at "cloudshell" in the home directory (~). They run "jobs" and see a list of three stopped jobs: [1] Stopped sleep 1000, [2]- Stopped sleep 1000, and [3]+ Stopped sleep 1000.

```
nishakar055@cloudshell:~$ jobs
[1]  Stopped                  sleep 1000
[2]-  Stopped                  sleep 1000
[3]+  Stopped                  sleep 1000
nishakar055@cloudshell:~$
```

54. Nice



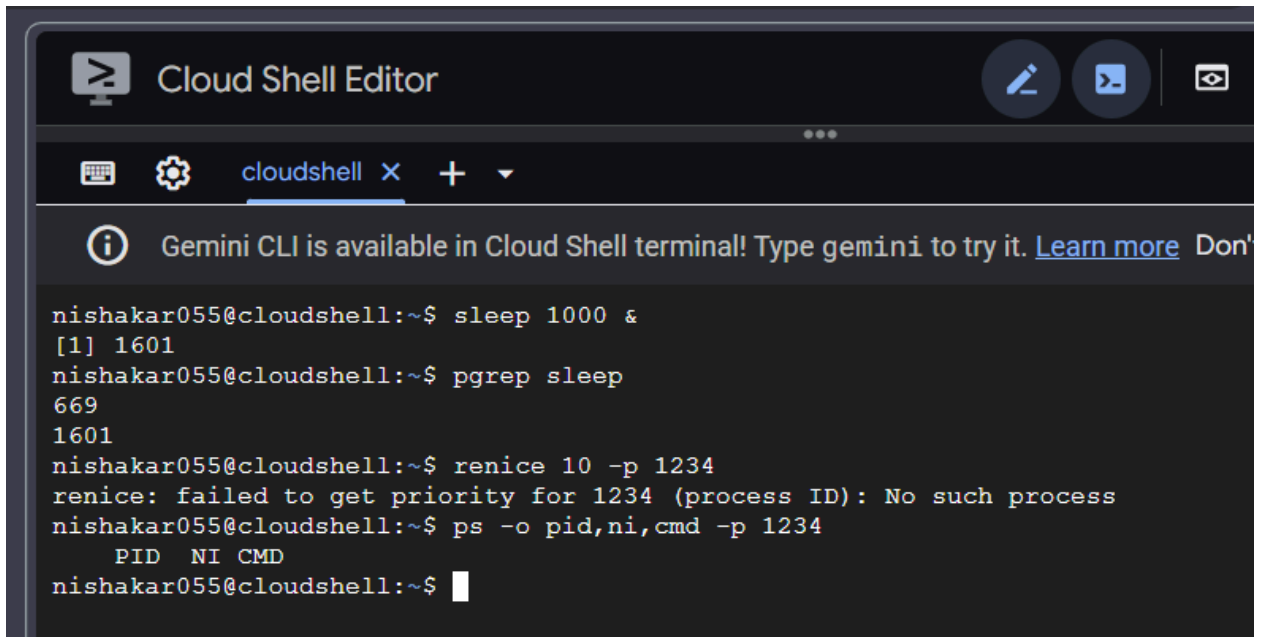
```
Cloud Shell Editor

cloudshell x + v

(i) Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't

nishakar055@cloudshell:~$ nice
-10
nishakar055@cloudshell:~$
```

55. Renice



```
Cloud Shell Editor

cloudshell x + v

(i) Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't

nishakar055@cloudshell:~$ sleep 1000 &
[1] 1601
nishakar055@cloudshell:~$ pgrep sleep
669
1601
nishakar055@cloudshell:~$ renice 10 -p 1234
renice: failed to get priority for 1234 (process ID): No such process
nishakar055@cloudshell:~$ ps -o pid,ni,cmd -p 1234
PID NI CMD
nishakar055@cloudshell:~$
```

56. Df

Cloud Shell Editor

cloudshell

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#) Don't show again

nishakar055@cloudshell:~\$ df

Filesystem	1K-blocks	Used	Available	Use%	Mounted on
overlay	98831908	76300060	22515464	78%	/
tmpfs	65536	0	65536	0%	/dev
/dev/sda1	98831908	76300060	22515464	78%	/root
/dev/disk/by-id/google-home-part1	5018272	156712	4583084	4%	/home
/dev/root	2019696	1288424	731272	64%	/usr/lib/modules
shm	65536	0	65536	0%	/dev/shm

nishakar055@cloudshell:~\$

57. Du

```
Cloud Shell Editor

cloudshell X + v

(i) Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't show again

8      ./codeoss/data/logs/20260515T032120/exthost1/vscode.github
196    ./codeoss/data/logs/20260515T032120/exthost1
204    ./codeoss/data/logs/20260515T032120
8      ./codeoss/data/logs/20260515T031825/exthost1
16     ./codeoss/data/logs/20260515T031825
8      ./codeoss/data/logs/20260515T040857/exthost1/vscode.git
444    ./codeoss/data/logs/20260515T040857/exthost1/output_logging_20260515T040917
8      ./codeoss/data/logs/20260515T040857/exthost1/vscode.github-authentication
4      ./codeoss/data/logs/20260515T040857/exthost1/vscode.json-language-features
8      ./codeoss/data/logs/20260515T040857/exthost1/vscode.github
480    ./codeoss/data/logs/20260515T040857/exthost1
488    ./codeoss/data/logs/20260515T040857
8      ./codeoss/data/logs/20260514T063428/exthost1/vscode.git
8      ./codeoss/data/logs/20260514T063428/exthost1/vscode.github-authentication
4      ./codeoss/data/logs/20260514T063428/exthost1/vscode.json-language-features
8      ./codeoss/data/logs/20260514T063428/exthost1/vscode.github
352    ./codeoss/data/logs/20260514T063428/exthost1/output_logging_20260514T063810
260    ./codeoss/data/logs/20260514T063428/exthost1/output_logging_20260514T063436
652    ./codeoss/data/logs/20260514T063428/exthost1
8      ./codeoss/data/logs/20260514T063428/exthost2/vscode.git
8      ./codeoss/data/logs/20260514T063428/exthost2/vscode.github-authentication
4      ./codeoss/data/logs/20260514T063428/exthost2/vscode.json-language-features
248    ./codeoss/data/logs/20260514T063428/exthost2/output_logging_20260514T070232
8      ./codeoss/data/logs/20260514T063428/exthost2/vscode.github
288    ./codeoss/data/logs/20260514T063428/exthost2
248    ./codeoss/data/logs/20260514T063428/exthost3/output_logging_20260514T072541
8      ./codeoss/data/logs/20260514T063428/exthost3/vscode.git
8      ./codeoss/data/logs/20260514T063428/exthost3/vscode.github-authentication
4      ./codeoss/data/logs/20260514T063428/exthost3/vscode.json-language-features
8      ./codeoss/data/logs/20260514T063428/exthost3/vscode.github
284    ./codeoss/data/logs/20260514T063428/exthost3
1232   ./codeoss/data/logs/20260514T063428
5060   ./codeoss/data/logs
1552   ./codeoss/data/CachedProfilesData/__default__profile__
1556   ./codeoss/data/CachedProfilesData
6652   ./codeoss/data
8      ./codeoss/extensions
6664   ./codeoss
156680 .
nishakar055@cloudshell:~$
```

58.fdisk


```
Cloud Shell Editor

cloudshell x + v

(i) Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't show again Dis

Disklabel type: gpt
Disk identifier: 88CD0047-6E3D-984A-9279-9D30D1B6BC8F

Device      Start      End      Sectors  Size Type
/dev/sda1   8704000    209715166 201011167 95.8G Linux filesystem
/dev/sda2    20480      53247     32768    16M ChromeOS kernel
/dev/sda3   4509696    8703999   4194304   2G ChromeOS root fs
/dev/sda4    53248      86015     32768    16M ChromeOS kernel
/dev/sda5   315392    4509695   4194304   2G ChromeOS root fs
/dev/sda6   16448      16448      1    512B ChromeOS kernel
/dev/sda7   16449      16449      1    512B ChromeOS root fs
/dev/sda8    86016     118783     32768    16M Linux filesystem
/dev/sda9    16450      16450      1    512B ChromeOS reserved
/dev/sda10   16451      16451      1    512B ChromeOS reserved
/dev/sda11     64      16447     16384     8M BIOS boot
/dev/sda12  249856    315391     65536    32M EFI System

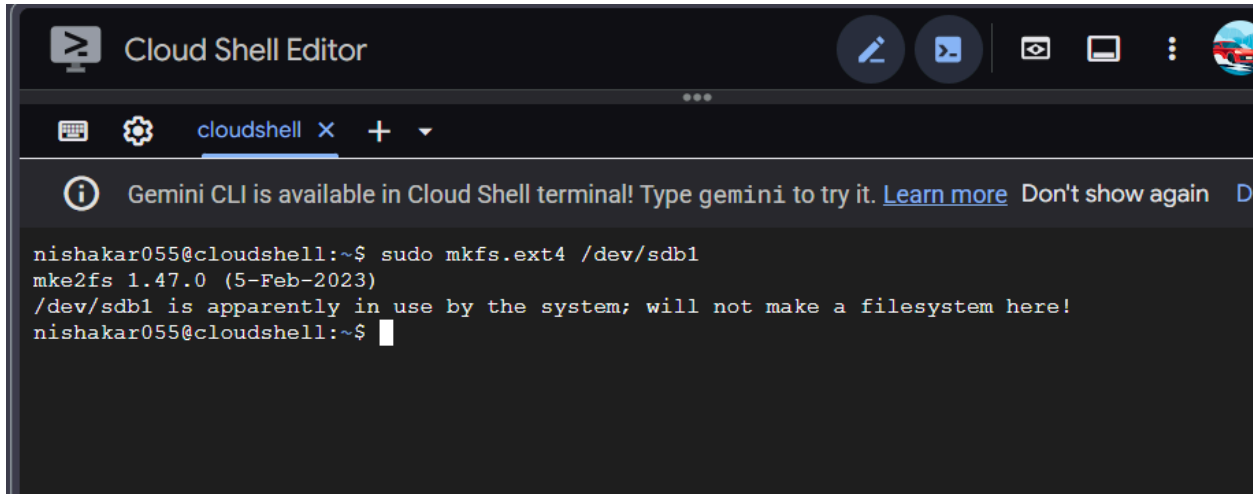
Partition 7 does not start on physical sector boundary.
Partition 9 does not start on physical sector boundary.
Partition 10 does not start on physical sector boundary.
Partition table entries are not in disk order.

Disk /dev/dm-0: 1.94 GiB, 2087714816 bytes, 509696 sectors
Units: sectors of 1 * 4096 = 4096 bytes
Sector size (logical/physical): 4096 bytes / 4096 bytes
I/O size (minimum/optimal): 4096 bytes / 4096 bytes

Disk /dev/sdb: 5 GiB, 5368709120 bytes, 10485760 sectors
Disk model: PersistentDisk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 4096 bytes
I/O size (minimum/optimal): 4096 bytes / 4096 bytes
Disklabel type: dos
Disk identifier: 0x5245a676

Device      Boot Start      End      Sectors  Size Id Type
/dev/sdb1    2048 10485759 10483712    5G 83 Linux
nishakar055@cloudshell:~$
```

59. Mkfs



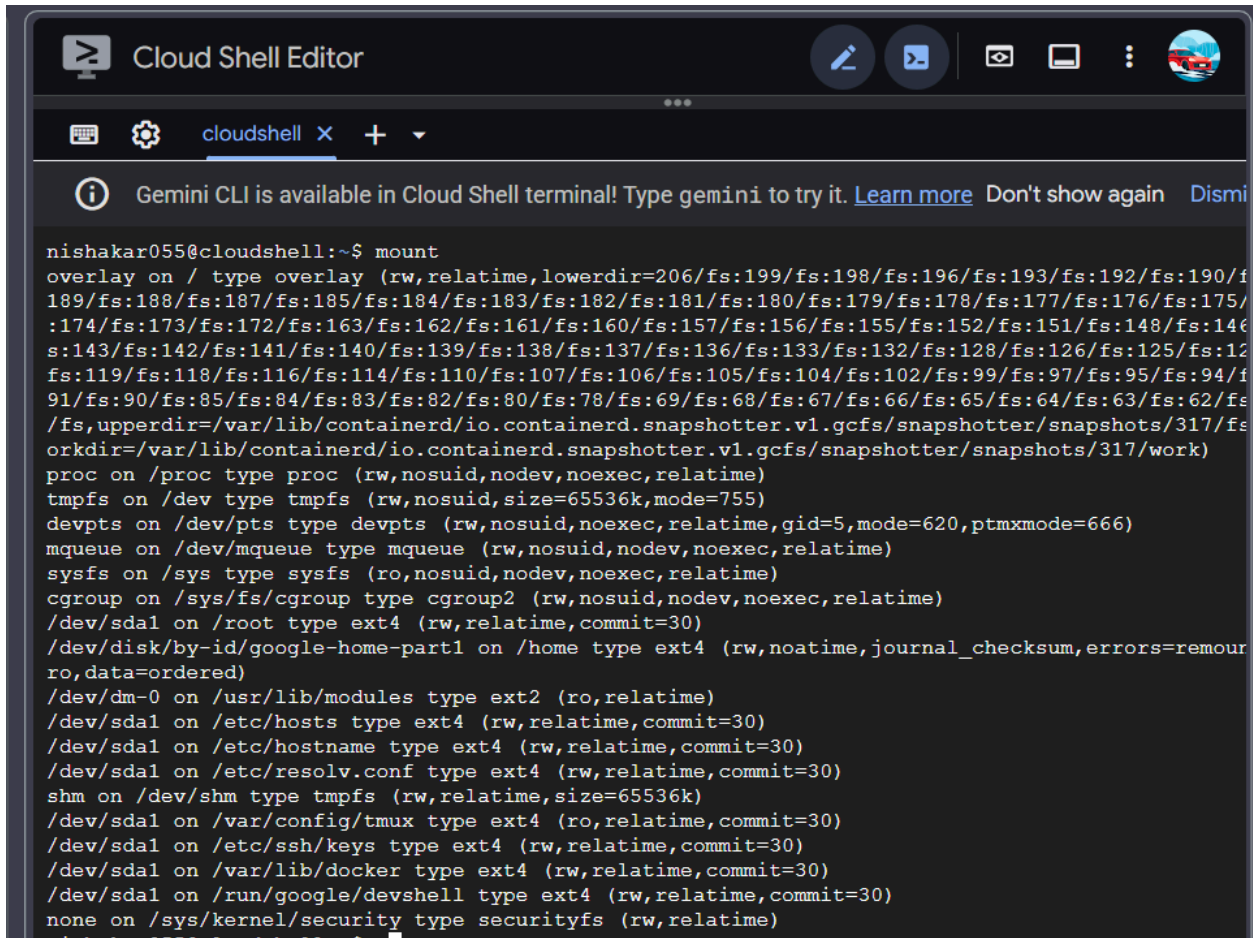
```
Cloud Shell Editor

cloudshell x + v

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't show again D

nishakar055@cloudshell:~$ sudo mkfs.ext4 /dev/sdb1
mkfs2fs 1.47.0 (5-Feb-2023)
/dev/sdb1 is apparently in use by the system; will not make a filesystem here!
nishakar055@cloudshell:~$
```

60.mount



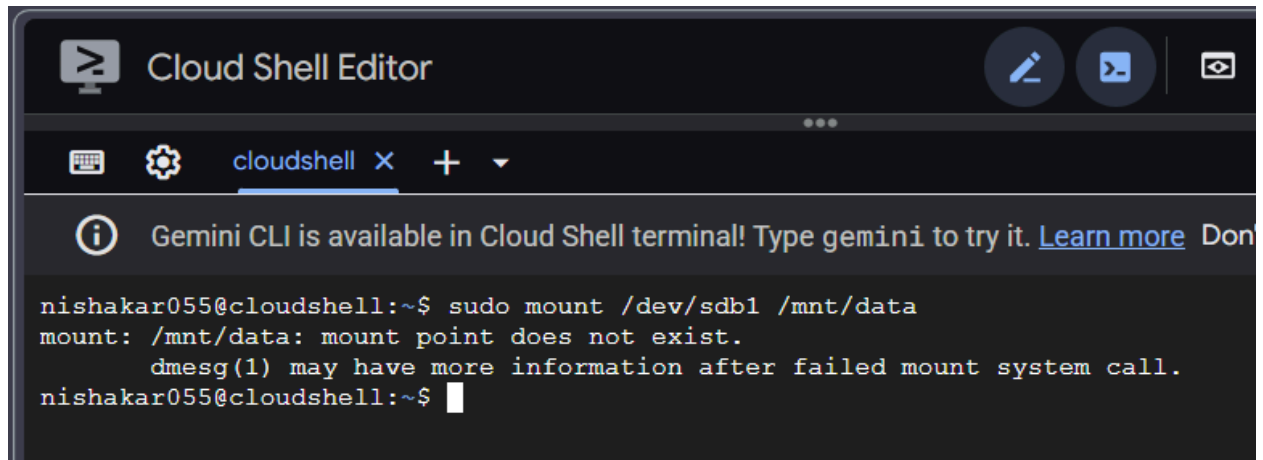
```
Cloud Shell Editor

cloudshell x + v

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't show again Dismiss

nishakar055@cloudshell:~$ mount
overlay on / type overlay (rw,relatime,lowerdir=206/fs:199/fs:198/fs:196/fs:193/fs:192/fs:190/fs:189/fs:188/fs:187/fs:185/fs:184/fs:183/fs:182/fs:181/fs:180/fs:179/fs:178/fs:177/fs:176/fs:175/fs:174/fs:173/fs:172/fs:163/fs:162/fs:161/fs:160/fs:157/fs:156/fs:155/fs:152/fs:151/fs:148/fs:146/fs:143/fs:142/fs:141/fs:140/fs:139/fs:138/fs:137/fs:136/fs:133/fs:132/fs:128/fs:126/fs:125/fs:124/fs:119/fs:118/fs:116/fs:114/fs:110/fs:107/fs:106/fs:105/fs:104/fs:102/fs:99/fs:97/fs:95/fs:94/fs:91/fs:90/fs:85/fs:84/fs:83/fs:82/fs:80/fs:78/fs:69/fs:68/fs:67/fs:66/fs:65/fs:64/fs:63/fs:62/fs:61/fs,upperdir=/var/lib/containerd/io.containerd.snapshotter.v1.gcfs/snapshotter/snapshots/317/fs,workdir=/var/lib/containerd/io.containerd.snapshotter.v1.gcfs/snapshotter/snapshots/317/work)
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
tmpfs on /dev type tmpfs (rw,nosuid,size=65536k,mode=755)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,gid=5,mode=620,ptmxmode=666)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime)
sysfs on /sys type sysfs (ro,nosuid,nodev,noexec,relatime)
cgroup on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime)
/dev/sda1 on /root type ext4 (rw,relatime,commit=30)
/dev/disk/by-id/google-home-part1 on /home type ext4 (rw,noatime,journal_checksum,errors=remount-ro,data=ordered)
/dev/dm-0 on /usr/lib/modules type ext2 (ro,relatime)
/dev/sda1 on /etc/hosts type ext4 (rw,relatime,commit=30)
/dev/sda1 on /etc/hostname type ext4 (rw,relatime,commit=30)
/dev/sda1 on /etc/resolv.conf type ext4 (rw,relatime,commit=30)
shm on /dev/shm type tmpfs (rw,relatime,size=65536k)
/dev/sda1 on /var/config/tmux type ext4 (ro,relatime,commit=30)
/dev/sda1 on /etc/ssh/keys type ext4 (rw,relatime,commit=30)
/dev/sda1 on /var/lib/docker type ext4 (rw,relatime,commit=30)
/dev/sda1 on /run/google/devshell type ext4 (rw,relatime,commit=30)
none on /sys/kernel/security type securityfs (rw,relatime)
nishakar055@cloudshell:~$
```

61.umount



The screenshot shows the Cloud Shell Editor interface. At the top, there's a title bar with a terminal icon and the text "Cloud Shell Editor". Below this is a tab bar with a keyboard icon, a gear icon, and a tab labeled "cloudshell" with a close button and a plus sign. A notification banner at the top of the terminal area says "Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#) Don". The terminal itself shows a user prompt "nishakar055@cloudshell:~\$" followed by the command "sudo mount /dev/sdb1 /mnt/data". The output is "mount: /mnt/data: mount point does not exist." followed by "dmesg(1) may have more information after failed mount system call." and another user prompt "nishakar055@cloudshell:~\$" with a cursor.

```
nishakar055@cloudshell:~$ sudo mount /dev/sdb1 /mnt/data
mount: /mnt/data: mount point does not exist.
       dmesg(1) may have more information after failed mount system call.
nishakar055@cloudshell:~$
```

62. Blkid

```
Cloud Shell Editor

cloudshell x + v

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't show

Disklabel type: gpt
Disk identifier: 88CD0047-6E3D-984A-9279-9D30D1B6BC8F

Device      Start      End      Sectors  Size Type
/dev/sda1    8704000    209715166 201011167 95.8G Linux filesystem
/dev/sda2      20480      53247      32768    16M ChromeOS kernel
/dev/sda3    4509696    8703999    4194304    2G ChromeOS root fs
/dev/sda4      53248      86015      32768    16M ChromeOS kernel
/dev/sda5    315392    4509695    4194304    2G ChromeOS root fs
/dev/sda6     16448      16448        1    512B ChromeOS kernel
/dev/sda7     16449      16449        1    512B ChromeOS root fs
/dev/sda8     86016     118783      32768    16M Linux filesystem
/dev/sda9     16450      16450        1    512B ChromeOS reserved
/dev/sda10    16451      16451        1    512B ChromeOS reserved
/dev/sda11        64      16447      16384     8M BIOS boot
/dev/sda12   249856    315391      65536    32M EFI System

Partition 7 does not start on physical sector boundary.
Partition 9 does not start on physical sector boundary.
Partition 10 does not start on physical sector boundary.
Partition table entries are not in disk order.

Disk /dev/dm-0: 1.94 GiB, 2087714816 bytes, 509696 sectors
Units: sectors of 1 * 4096 = 4096 bytes
Sector size (logical/physical): 4096 bytes / 4096 bytes
I/O size (minimum/optimal): 4096 bytes / 4096 bytes

Disk /dev/sdb: 5 GiB, 5368709120 bytes, 10485760 sectors
Disk model: PersistentDisk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 4096 bytes
I/O size (minimum/optimal): 4096 bytes / 4096 bytes
Disklabel type: dos
Disk identifier: 0x5245a676

Device      Boot Start      End  Sectors  Size Id Type
/dev/sdb1    2048 10485759 10483712    5G 83 Linux
nishakar055@cloudshell:~$
```

63. Lsblk

 Cloud Shell Editor

cloudshell X + ▾



 Gemini CLI is available in Cloud Shell terminal! Type gemini

```
nishakar055@cloudshell:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0      0  100G  0 disk
├─sda1       8:1      0  95.8G  0 part /run/google/devshell
│            │
│            │ /var/lib/docker
│            │ /etc/ssh/keys
│            │ /var/config/tmux
│            │ /etc/resolv.conf
│            │ /etc/hostname
│            │ /etc/hosts
│            │ /root
├─sda2       8:2      0   16M  0 part
├─sda3       8:3      0    2G  0 part
├─sda4       8:4      0   16M  0 part
├─sda5       8:5      0    2G  0 part
├─sda6       8:6      0   512B  0 part
├─sda7       8:7      0   512B  0 part
├─sda8       8:8      0   16M  0 part
├─sda9       8:9      0   512B  0 part
├─sda10      8:10     0   512B  0 part
├─sda11      8:11     0    8M  0 part
└─sda12      8:12     0   32M  0 part
sdb          8:16     0    5G  0 disk
└─sdb1       8:17     0    5G  0 part /home
nishakar055@cloudshell:~$
```

64. parted

```
Cloud Shell Editor

cloudshell x + v

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't sho

nishakar055@cloudshell:~$ sudo parted -l
Model: Google PersistentDisk (scsi)
Disk /dev/sda: 107GB
Sector size (logical/physical): 512B/4096B
Partition Table: gpt
Disk Flags: pmbr_boot

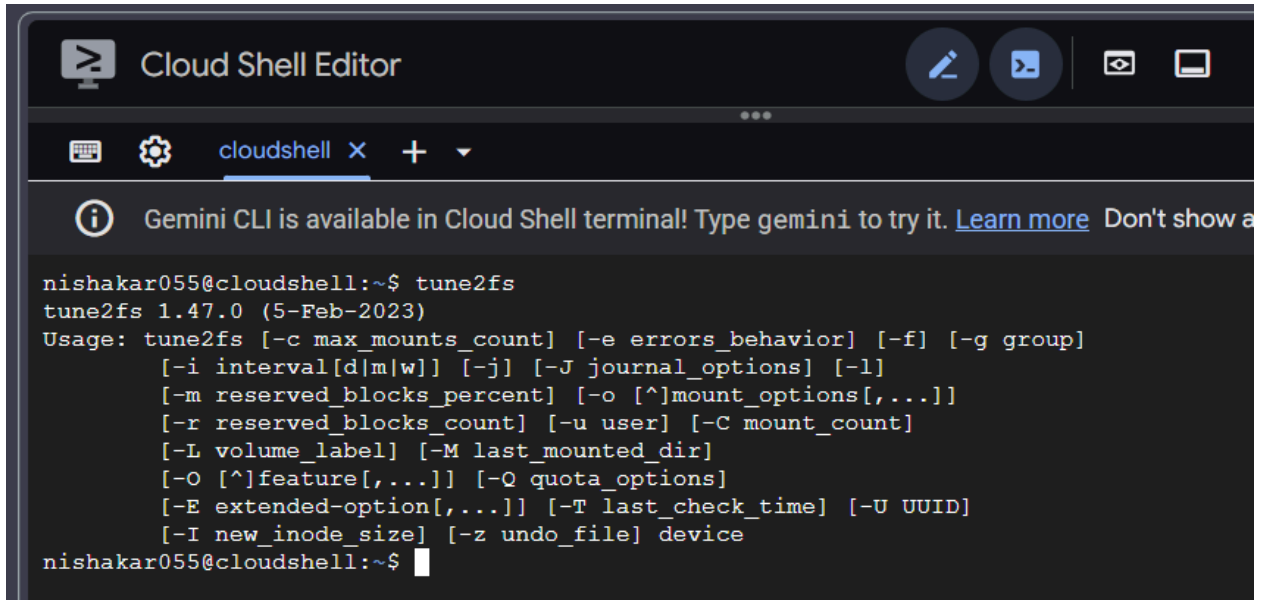
Number  Start   End     Size    File system  Name        Flags
11      32.8kB  8421kB  8389kB  RWFW         bios_grub   bios_grub
6       8421kB  8422kB  512B    KERN-C       chromeos_kernel
7       8422kB  8422kB  512B    ROOT-C       chromeos_kernel
9       8422kB  8423kB  512B    reserved
10      8423kB  8423kB  512B    reserved
2       10.5MB  27.3MB  16.8MB  KERN-A       chromeos_kernel
4       27.3MB  44.0MB  16.8MB  KERN-B       chromeos_kernel
8       44.0MB  60.8MB  16.8MB  ext4         OEM
12      128MB   161MB   33.6MB  fat16        EFI-SYSTEM  boot, legacy_boot, esp
5       161MB   2309MB  2147MB  ROOT-B
3       2309MB  4456MB  2147MB  ext2         ROOT-A
1       4456MB  107GB   103GB   ext4         STATE

Model: Google PersistentDisk (scsi)
Disk /dev/sdb: 5369MB
Sector size (logical/physical): 512B/4096B
Partition Table: msdos
Disk Flags:

Number  Start   End     Size    Type    File system  Flags
1       1049kB  5369MB  5368MB  primary ext4

nishakar055@cloudshell:~$
```

65. tune2fs

A screenshot of a Cloud Shell Editor terminal window. The window has a dark theme. At the top, there's a title bar with a terminal icon and the text "Cloud Shell Editor". Below the title bar, there's a tab labeled "cloudshell" with a close button and a plus sign. A notification bar at the top of the terminal area says "Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#) Don't show a". The terminal content shows a user running the "tune2fs" command. The output displays the version "1.47.0 (5-Feb-2023)" and a detailed usage message with various options like [-c max_mounts_count], [-e errors_behavior], [-f], [-g group], [-i interval[d|m|w]], [-j], [-J journal_options], [-l], [-m reserved_blocks_percent], [-o [^]mount_options[,...]], [-r reserved_blocks_count], [-u user], [-C mount_count], [-L volume_label], [-M last_mounted_dir], [-O [^]feature[,...]], [-Q quota_options], [-E extended-option[,...]], [-T last_check_time], [-U UUID], [-I new_inode_size], [-z undo_file], and device. The prompt "nishakar055@cloudshell:~\$" is visible at the bottom.

```
nishakar055@cloudshell:~$ tune2fs
tune2fs 1.47.0 (5-Feb-2023)
Usage: tune2fs [-c max_mounts_count] [-e errors_behavior] [-f] [-g group]
        [-i interval[d|m|w]] [-j] [-J journal_options] [-l]
        [-m reserved_blocks_percent] [-o [^]mount_options[,...]]
        [-r reserved_blocks_count] [-u user] [-C mount_count]
        [-L volume_label] [-M last_mounted_dir]
        [-O [^]feature[,...]] [-Q quota_options]
        [-E extended-option[,...]] [-T last_check_time] [-U UUID]
        [-I new_inode_size] [-z undo_file] device
nishakar055@cloudshell:~$
```

66. lostat

```
Cloud Shell Editor

cloudshell x + v

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't show again Dismiss

0 upgraded, 1 newly installed, 0 to remove and 2 not upgraded.
Need to get 489 kB of archives.
After this operation, 1,487 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 sysstat amd64 12.6.1-2 [489 kB]
Fetched 489 kB in 0s (8,724 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package sysstat.
(Reading database ... 121230 files and directories currently installed.)
Preparing to unpack .../sysstat_12.6.1-2_amd64.deb ...
Unpacking sysstat (12.6.1-2) ...
Setting up sysstat (12.6.1-2) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used)
. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 79.)
debconf: falling back to frontend: Readline

Creating config file /etc/default/sysstat with new version
update-alternatives: using /usr/bin/sar.sysstat to provide /usr/bin/sar (sar) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/sar.1.gz because associated
file /usr/share/man/man1/sar.sysstat.1.gz (of link group sar) doesn't exist
Created symlink /etc/systemd/system/sysstat.service.wants/sysstat-collect.timer → /usr/lib/systemd/system/sysstat-collect.timer.
Created symlink /etc/systemd/system/sysstat.service.wants/sysstat-summary.timer → /usr/lib/systemd/system/sysstat-summary.timer.
Created symlink /etc/systemd/system/multi-user.target.wants/sysstat.service → /usr/lib/systemd/system/sysstat.service.
Processing triggers for man-db (2.12.0-4build2) ...
nishakar055@cloudshell:~$ iostat
Linux 6.6.137+ (cs-524432981010-default)      05/15/2026      _x86_64_      (4 CPU)

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           0.70    0.00   0.37   0.09   0.00   98.84

Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd
dm-0               1.45       79.06         0.00         0.00    434300         0         0
sda                6.99      308.18        46.57         0.00   1692883      255797         0
sdb                0.19       24.89         0.36         0.00    136749       1984         0

nishakar055@cloudshell:~$
```

67. Vmstat


```
Cloud Shell Editor

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't show again

nishakar055@cloudshell:~$ vmstat
procs -----memory----- ---swap-- -----io----- -system-- -----cpu-----
 r b   swpd   free   buff  cache   si   so    bi    bo    in   cs us  sy id wa st gu
  1  0       0 12350900 80468 1947144    0    0   331   47 793    3  1  0 99  0  0  0
nishakar055@cloudshell:~$
```

68. Mpstat

```
Cloud Shell Editor

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't show again Dismiss

nishakar055@cloudshell:~$ mpstat
Linux 6.6.137+ (cs-524432981010-default)      05/15/2026      _x86_64_      (4 CPU)

04:22:52 AM CPU      %usr   %nice    %sys %iowait    %irq   %soft  %steal  %guest  %gnice   %idle
04:22:52 AM all      0.70    0.00    0.34    0.09    0.00    0.02    0.00    0.00    0.00   98.86
nishakar055@cloudshell:~$
```